

Data-driven integrated home service staffing and capacity planning: Stochastic optimization approaches

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ABSTRACT

We address a *home service staffing and capacity planning* (HSCP) problem under uncertainty. Specifically, given sets of providers, service types, and days in the planning horizon, we aim to determine the number of providers to hire (*staffing*) and the allocation of hired providers to different types of services (*capacity planning*). Data from a collaborating home services provider in Beijing demonstrate significant variability in customer demand within and across service types. Service duration is also random. The objective is to minimize the total cost associated with staffing, capacity allocation, over-staffing, and under-staffing. We propose two-stage stochastic programming (SP) and data-driven distributionally robust optimization (DRO) approaches to address demand and service time uncertainty considering two types of decision-makers, namely an everything-in-advance decision-maker (EA) and a flexible adjustment decision-maker (FA). In the EA models, we determine the staffing and capacity allocation decisions in the first stage before observing the demand. In the FA models, we make staffing decisions in the first stage and then determine the allocations based on demand realizations in the second stage. We derive equivalent mixed-integer linear programming (MILP) reformulations of the proposed DRO models for the EA decision-maker that can be implemented and efficiently solved using off-the-shelf optimization software. We propose a computationally efficient column-and-constraint generation algorithm with valid inequalities to solve the proposed DRO models for the FA decision-maker. Numerical experiments based on data from a home services provider in Beijing are used to compare the proposed approaches and illustrate the potential for impact in practice.

1. Introduction

Home service is the provision of providing services to people at their homes, such as home care and home banking. The home service industry has been rapidly growing due to the aging population, outspread of infectious diseases, emergent changes in family structures, work obligations, and extended work hours. For example, the recent COVID-19 pandemic has led to unprecedented increases in the demand for home healthcare to support vulnerable and high-risk populations (Alkaabneh and Diabat, 2023; Tsang and Shehadeh, 2023).

The demand for home service is associated with increased demand for cross-trained providers who can provide multiple service types. Unfortunately, the home service industry faces a significant shortage of providers. For example, Washington (2018) recently reported that labor shortages in Washington State would worsen in the next twenty years. Moreover, the state estimates that nearly 77,000 additional home providers will be needed by 2030. The home service costs are also increasing. According to the newly released Genworth cost of care

survey (Genworth, 2022), the cost of home health providers, who provide personal assistance with activities such as bathing, dressing and eating, grew by 12.5% in 2020. Accordingly, the development of computationally efficient optimization tools for home service staffing and capacity allocation is essential to support decision-making in the home service industry.

In this paper, we study a staffing and capacity planning problem arising from home service practice. Specifically, we consider the perspective of a home service agency that must decide the number of providers to hire (*staffing*) within a specified planning horizon and the allocation of hired providers to different types of services in each day in the planning horizon (*capacity allocation*). Customers demand and durations for different types of services are random. The objective is to find staffing and capacity allocation decisions that minimize total employment cost, capacity allocation cost, and penalty cost for daily over-staffing and under-staffing. We call this problem the *home service staffing and capacity planning* (HSCP) problem.

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The HSCP problem is a challenging optimization problem for several reasons. First, it is a complex combinatorial optimization problem that requires deciding the number of different (possibly cross-trained) providers to be hired and allocating (i.e., assigning) the capacity of hired providers to different types of services. Second, there is significant variability in demand for home services and service duration. In addition, our analysis of real-life data from a home service provider suggest that there is a wide range of possible probability distributions for modeling the variability in the demand, indicating distributional ambiguity, i.e., uncertainty in distribution type (see this analysis in Section 2). Such uncertainty and distributional ambiguity increase the complexity of modeling and solving the HSCP problem. However, ignoring uncertainty may lead to sub-optimal decisions. Notably, such decisions may lead to the inability to meet customer demand and high costs, among other disappointing consequences. Failure to meet customer demand in healthcare, for example, may impact population health. It also impacts customers' satisfaction and thus the reputation of the home service providers.

Motivated by these important issues, we propose new stochastic optimization models for modeling and solving the HSCP problem under uncertainty. Based on the practice of a large home care provider in Beijing, we consider two types of decision-makers, namely *Everything in Advance* (EA) and *Flexible Adjustment* (FA) decision-makers. These decision-makers have different planning and decision-making attitude under uncertainty. The EA decision-makers prefer to plan all decisions in advance and serve customers according to the established plans. Hence, they decide the number of providers to hire and their daily capacity allocation to service types at the start of the planning horizon before the demand and service time are realized. In contrast, the FA decision-makers adopt flexible capacity allocation to hedge against uncertainty. Specifically, they decide the number of providers to hire at the beginning of the planning horizon. Then, they allocate the hired providers to service types based on demand realizations.

To address uncertainty, we propose two-stage stochastic programming (SP) and distributionally robust optimization (DRO) models for both decision-makers. In the SP approach, we assume that decision-makers know the distributions of demand and service duration. In practice and as our data suggest, the data-generating probability distribution remains unobservable to the decision-maker. Therefore, to address potential distributional ambiguity, we propose DRO models. In these models, we model the uncertainty distribution as a decision variable that belongs to an ambiguity set (i.e., a set of distributions). Then, the optimization is performed based on distribution within this set. We employ two popular types of ambiguity sets in our DRO models. The first is a data-driven 1-Wasserstein ambiguity set, which allows for incorporating possibly small-size data in the ambiguity set and optimization. The second, is a mean-support ambiguity set that incorporates estimates of the demand and service duration's mean and support (range). Thus, this ambiguity set is based on intuitive statistics that decision-makers may conveniently approximate or estimate.

We derive mixed-integer linear programming reformulations of our proposed DRO models for the EA decision-maker that can be implemented and efficiently solved using off-the-shelf optimization software. We propose a computationally efficient column-and-constraint generation (C&CG) method to solve our proposed DRO model for the FA decision-maker. In addition, we derive valid inequalities to accelerate C&CG's convergence. Finally, using data from a home services provider in Beijing and parameter settings from the literature, we generate various problem instances to investigate the proposed models' computational and operational performances. Additionally, we derive several managerial insights relevant to practice. To the best of our knowledge, and according to our literature review in Section 3, our paper is the first to propose stochastic optimization methodologies for modeling and solving the HSCP problem under uncertainty considering both EA and FA decision-makers. The goal is not to advocate for any of these different approaches of modeling uncertainty in the HSCP but rather to investigate their potential for impact on practice.

1.1. Structure of the paper

The remainder of this paper is organized as follows. In Section 2, we present a motivating example for modeling uncertainty and distributional ambiguity. In Section 3, we review the relevant literature. In Section 4 we formally define the HSCP problem. In Sections 4.1–4.2, we present our proposed SP and DRO models. In Section 5, we present our C&CG algorithm and strategies to speed convergence. In Section 6, we present our computational results and managerial implications. Finally, we draw conclusions in Section 7.

2. Motivation

We obtain data from a leading home service providers in Beijing, China (Pinetree Care). This company provides more than 5 million care sessions to over 600,000 families through a network of more than 10,000 providers. One of the main challenges that this company and other home services companies face when making staffing and capacity allocation decisions is the uncertainty in the demand for service and service duration. To analyze this uncertainty, we use data from Pinetree Care collected electronically from June 2017 through December 2019, representing ~205K care sessions in Beijing.

In the top panel of Fig. 1, we present the empirical and fitted probability distributions of the data on the actual demand for rehabilitation nursing (RN) and health status assessment (HSA) services. In Appendix A we present the associated goodness-of-fit metrics (e.g., Negative of the Log-Likelihood (NLogL), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC)). In the bottom panel of Fig. 1, we present demand distributions during the fourth quarter of 2017 and the fourth quarter of 2018.

We make the following observations from these results. First, Figs. 1(a)–1(b) show significant variability in demand within and across service types. Furthermore, these figures, along with goodness-of-fit metrics, demonstrate that a wide range of distributions could represent uncertainty in demand, suggesting distributional ambiguity. Figs. 1(c)–1(d) show that demand variability and distributions are quite different between the fourth quarters of 2017 and 2018 (see also Figure A.1 in Appendix A). Indeed, future uncertainty is often not distributed as in the past. Finally, these results motivate us to propose DRO approaches for the HSCP problem to obtain robust decisions that could hedge against potential distributional ambiguity.

3. Literature review

In this section, we focus primarily on the literature related to us, i.e., papers that apply stochastic optimization to address problems similar to the HSCP problem. For comprehensive surveys of operations research methods applied to home care problems, we refer to Gutiérrez and Vidal (2013), Fikar and Hirsch (2017), Grieco et al. (2021) and Di Mascolo et al. (2021).

Existing stochastic home service staffing and capacity planning models often consider demand uncertainty and employ the SP approach. SP approaches for problems similar to the HSCP include Rodríguez et al. (2015), and Zheng et al. (2021). Rodríguez et al. (2015) proposed a two-stage SP approach to solve the staff dimensioning problem in home care services considering random demand. The first stage finds levels of resources to use, while the second stage computes the optimal number of provider for each profession. The recent work of Zheng et al. (2021) is closely related to ours. Specifically, Zheng et al. (2021) considered a multi-resource and multi-demand network matching problem in which they assume deterministic service duration and that the demands for services follow a fully known probability distribution. Accordingly, Zheng et al. (2021) proposed a two-stage SP which determine the service authorization and capacity planning decision in first stage, and resource allocation to services in the second stage. The goal is to maximize the total expected revenue netted by the

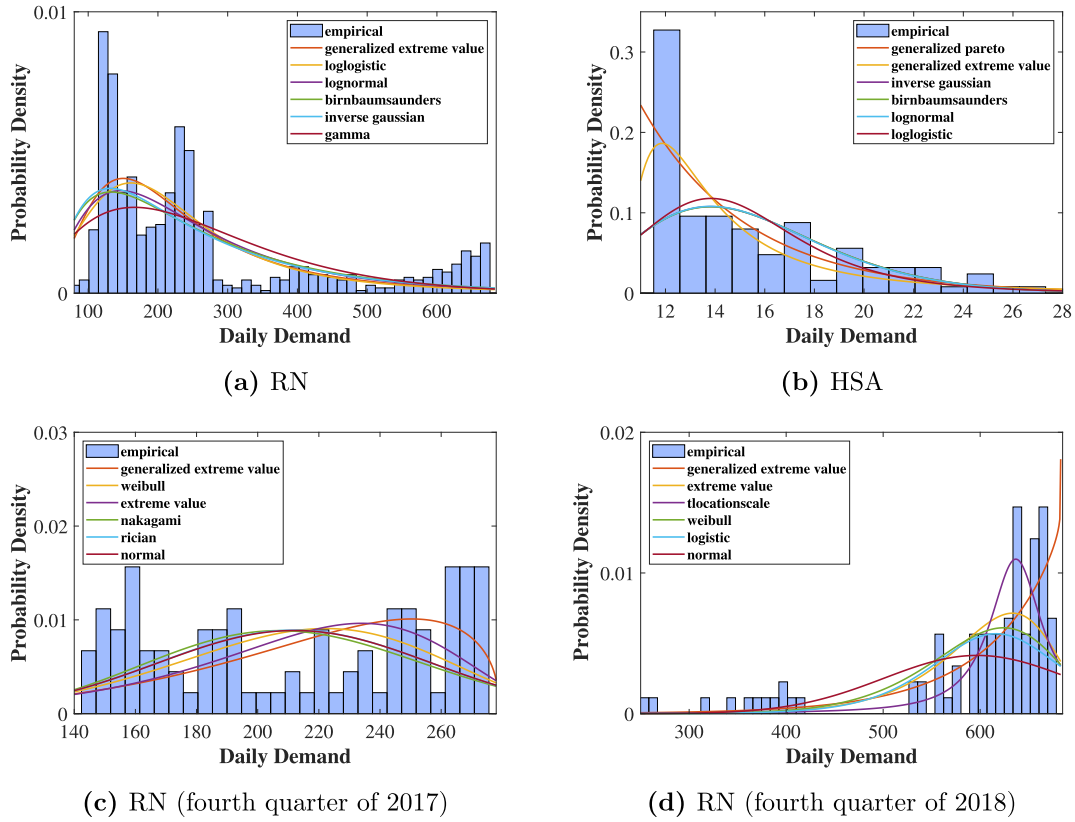


Fig. 1. The empirical and fitted probability distributions for actual demands.

authorization cost, payment and the penalty cost of service shortage. However, they do not incorporate the penalty cost of over-staffing in the objective, which is an important source of operational expenses. In addition, ignoring random service duration may lead to sub-optimal solutions. Our models incorporate these elements, among others. Finally, we remark that while some studies considered FA decision-makers, to the best of our knowledge and according to recent surveys, no one has yet developed optimization models for EA decision-makers in the HSCP context.

The applicability of the SP approach is limited to the case in which the decision-makers have sufficient data to model the distribution of random parameters. Moreover, optimal solutions to SP models are known to be sensitive to the distribution type (or sample) employed in the optimization, and hence, may show disappointing out-of-sample performance under the true distribution or unseen data (Mohajerin Esfahani and Kuhn, 2018).

Robust optimization (RO) is an alternative technique to model, analyze and optimize decisions under uncertainty and ambiguity (where the distributions are unknown). In RO, one first defines an *uncertainty set* in which the uncertain parameters lie; see, e.g., Ben-Tal et al. (2009), Bertsimas and Sim (2004). Decisions are then optimized against the worst-case realization of uncertain parameters within the uncertainty set. In home service area, several studies propose RO approaches to address the uncertainty (see, e.g., Carello and Lanzarone, 2014; Cappanera et al., 2018; Shi et al., 2019; Cappanera and Scutellà, 2021). Cappanera et al. (2018) addressed uncertainty of patient demand over a multiple-day time horizon and jointly studied the assignment, scheduling and routing decisions via a non-standard cardinality-constrained robust approach. Shi et al. (2019) proposed a robust model for both uncertain service and travel time and discuss the heuristic solution approaches to find the optimal route for each provider and appointment time for each patient. Although RO is a powerful technique in dealing with uncertainty in optimization, its solutions can be too conservative (Thiele, 2010; Roos and den Hertog, 2020).

Additionally, RO does not fully utilize any distributional information of uncertainty and focus the optimization on the worst-case scenario within a pre-defined *uncertainty set*. Hence, RO may yield poor expected performance and sub-optimal decisions for other more-likely scenarios. We refer to Bertsimas et al. (2011), Gabrel et al. (2014) and Gorissen et al. (2015) for comprehensive surveys on RO.

Distributionally robust optimization is another approach for modeling uncertainty. In DRO, we model the distribution of uncertainty as a decision variable that belongs to an ambiguity set (i.e., a family of distributions characterized through certain known properties of the unknown data-generating distribution, Rahimian and Mehrotra, 2019). Then, we optimize based on the worst-case distribution within the pre-defined ambiguity set. The DRO approach has the following benefits. First, it adopts a worst-case approach regularizing the optimization problem and thereby mitigating the optimizer's curse. Second, DRO avoids the well-known over-conservatism and the poor expected performance of RO. We refer to Rahimian and Mehrotra (2019) for a comprehensive survey on DRO techniques. We observe that despite the potential advantages, there is no DRO approaches for the specific HSCP problem that we address in this paper.

The computational tractability of a DRO model depends on the ambiguity set employed, and there are several popular types of ambiguity sets, including moment-based (Goh and Sim, 2010; Delage and Ye, 2010; Mehrotra and Zhang, 2014) and distance-based (Mohajerin Esfahani and Kuhn, 2018; Jiang et al., 2019) ambiguity sets. Most applied DRO literature employs moment-based ambiguity, which incorporates all distributions sharing certain moments (e.g., mean and support, see, e.g., Bertsimas and Popescu, 2005; Bertsimas et al., 2010; Delage and Ye, 2010; Jiang et al., 2017). This is because moment-based ambiguity sets can be conveniently calibrated using easy-to-compute or approximate information such as mean and range of random parameters. Moreover, moment-based DRO models often do not scale with the number of available uncertainty scenarios. Finally, various techniques

have been proposed to derive tractable reformulation of moment-based DRO models (Delage and Ye, 2010; Rahimian and Mehrotra, 2019).

Recent DRO approaches have shifted toward using distance metrics (e.g., ϕ -divergence Jiang and Guan, 2016, Wasserstein distance Mohajerin Esfahani and Kuhn, 2018; Gao and Kleywegt, 2016) to describe the deviation from a reference distribution. In this paper, we employ the 1-Wasserstein distance-based ambiguity set, which allows the incorporation of small-sized data in the ambiguity set and optimization. In addition, it enjoys asymptotic properties and often offers a strong out-of-sample performance guarantee (Mohajerin Esfahani and Kuhn, 2018). Moreover, it is richer than other divergence-based ambiguity sets. However, DRO models based on the Wasserstein distance scale with the data size.

In this paper, we propose two-stage SP models for the HSCP problem. In addition, we propose DRO models for the HSCP problem based on mean-support and 1-Wasserstein ambiguity sets to address the potential distributional ambiguity of the demand and service time. We compare the computational performance and solutions of the DRO models with their proposed SP counterparts. In contrast to Zheng et al. (2021), we address the co-existing uncertainty of demand and service time (recall that Zheng et al. (2021) assume that the service time is deterministic). Moreover, we propose SP and DRO models for two types of decision-makers (EA and FA). Finally, Zheng et al. (2021) assumed that the providers have single skill. In contrast, we additionally consider cross-trained providers.

4. Problem description and formulation

Let us now start to introduce our HSCP problem and proposed formulations. Suppose that there is a home service company that provides L different types of home services (e.g., home care, maintenance, home banking, etc.). The decision-maker of this company wants to determine the number of providers to hire within an arbitrary planning horizon of T days to provide home services to customers. Suppose that there are K types of providers to hire. Depending on their training, skills, and qualifications, each provider of type $k \in [K] := \{1, 2, \dots, K\}$, can provide a subset $R_k \subset [L]$ of services. In addition, the daily service capacity for type $k \in [K]$ providers is h_k . The fixed cost of hiring type k provider is c_k .

The demand, $d_{l,t}$, for service type $l \in [L]$ in each day $t \in [T]$ is random. The service time, $s_{l,t}$, for each service type $l \in [L]$ in each day $t \in [T]$ is also random. We assume that we know the mean and support of these random parameters. (Recall that these parameters can be estimated based on, e.g., expert knowledge, where support could represent the dispersion or range of variability that we seek protection against). We let $\mu_{l,t}^d$ and $\mu_{l,t}^s$ respectively represent the mean value of $d_{l,t}$ and $s_{l,t}$, for all $l \in [L]$ and $t \in [T]$. The random vector ξ is a measurable function $\xi: \Omega \rightarrow \mathcal{U}$ with a measurable space $(\Omega, \mathcal{F})$, where $\mathcal{U} = \mathcal{U}^d \times \mathcal{U}^s$ is the bounded support of ξ . $\mathcal{U}^d$ and $\mathcal{U}^s$ are respectively the support set of d and s defined as follows:

$$\mathcal{U}^d := \{d \geq 0 : \underline{d}_{l,t} \leq d_{l,t} \leq \bar{d}_{l,t}, \forall l \in [L], t \in [T]\}, \quad (1a)$$

$$\mathcal{U}^s := \{s \geq 0 : \underline{s}_{l,t} \leq s_{l,t} \leq \bar{s}_{l,t}, \forall l \in [L], t \in [T]\}. \quad (1b)$$

Due to the uncertainty of demand and service time, one or multiple of the following scenarios can happen, (1) over-staffing, or and, (2) under-staffing (i.e., failure to meet customers demand). The over-staffing and under-staffing unitary penalty costs are respectively $c_{l,t}^o$ and $c_{l,t}^u$, for all $l \in [L]$ and $t \in [T]$. A complete listing of the problem parameters can be found in Table 1.

Given T , K , and L , our models determine the number of providers to hire and the capacity of type k providers allocated to type $l \in [L]$ in each day $t \in [T]$. In particular, we consider two types of decision-makers. The first decides the number of providers to hire of each type and their daily capacity allocation to service types at the beginning of the planning horizon before the demand and service time are realized.

We call this decision-maker *Everything in Advance* (EA) decision-maker. The second decision-maker decides the number of providers to hire before realizing the demand (i.e., at the beginning of the planning horizon). S/he then allocates these providers to service types once the demand is realized. We call this decision-maker *Flexible Adjustment* (FA) decision-maker.

In the next sections, we propose two-stage SP and DRO models for EA and FA, considering the case of known and unknown distributions of demand and service time, respectively. In the first stage of the EA model, we decide the number of providers to hire from each type $k \in [K]$ and their capacity allocation to service type $l \in [L]$ in each day $t \in [T]$. And in the second stage, after the demand and service time are realized, we compute over-staffing and under-staffing. The objective is to minimize the total costs: fixed staffing (or hiring) and allocation (assignment) costs and a weighted sum of random costs related to over-staffing and under-staffing. In the first stage of the FA model, we decide the number of providers to hire, and in the second stage we decide capacity allocation and compute over-staffing and under-staffing. The objective of this model is to minimize the total costs: fixed hiring cost and a weighted sum of the random cost related to capacity allocation (assignment) cost, over-staffing cost, and under-staffing cost.

Additional notation: For $a \in \mathbb{Z}$, we define $[a] := \{1, 2, \dots, a\}$. For $a, b \in \mathbb{Z}$, $U[a; b]$ refers to uniform distribution over the interval $[a; b]$. We use boldface notation to denote vectors, e.g., $d := [d_1, d_2, \dots, d_N]^T$. Table 1 summarizes other notations.

4.1. Models for the EA decision-maker

4.1.1. Two-stage SP for EA

In this section, we present our proposed SP formulation of the HSCP problem considering the EA decision-maker. In this model (denoted as E-SP), we assume that we know the distribution $\mathbb{P}$ of uncertain parameters $\xi := [d, s]^T$. First, let us define the variables defining our E-SP model. Let integer decision variables x_k represent the number of type k providers hired, for all $k \in [K]$. Let continuous decision variables $y_{k,l,t}$ represent the capacity of type k providers allocated for type l service in day t , for all $k \in [K], l \in [L], t \in [T]$. Let continuous decision variables $o_{l,t}$ and $u_{l,t}$ represent over-staffing and under-staffing for type l service in each day t , for all $l \in [L], t \in [T]$. Our E-SP model can now be stated as follows:

$$(E-SP) \min_{x,y} \left\{ \sum_{k=1}^K c_k x_k + \sum_{k=1}^K \sum_{l=1}^L \sum_{t=1}^T c_{k,l,t} y_{k,l,t} + \mathbb{E}_{\mathbb{P}} [Q(x, y, \xi)] \right\} \quad (2a)$$

$$\text{s.t. } \underline{w} \leq \sum_{k=1}^K x_k \leq \bar{w}, \quad (2b)$$

$$\sum_{l \in R_k} y_{k,l,t} \leq x_k h_k, \quad \forall k \in [K], t \in [T], \quad (2c)$$

$$x_k \in \mathbb{Z}_+, y_{k,l,t} \in \mathbb{R}_+, \quad \forall k \in [K], l \in [L], t \in [T], \quad (2d)$$

where for each feasible decision x, y and a realization of $\xi := [d, s]^T$, our second-stage formulation is as follows:

$$Q(x, y, \xi) := \min_{o,u} \sum_{t=1}^T \sum_{l=1}^L (c_{l,t}^o o_{l,t} + c_{l,t}^u u_{l,t}) \quad (3a)$$

$$\text{s.t. } \sum_{k: l \in R_k} y_{k,l,t} + u_{l,t} - o_{l,t} = d_{l,t} s_{l,t}, \quad \forall l \in [L], t \in [T], \quad (3b)$$

$$o_{l,t}, u_{l,t} \in \mathbb{R}_+, \quad \forall l \in [L], t \in [T]. \quad (3c)$$

Formulation (2) seeks to find staffing and allocation decisions (x, y) that minimize the sum of (1) fixed employment and assignment costs (first two terms), and (2) expectation of the second stage (recourse)

Table 1

Notation.

Indices	
l	index of service types, $l \in [L]$
t	index of days, $t \in [T]$
k	index of providers types, $k \in [K]$
Parameters and sets	
L	number of service types
T	number of days in the planning horizon
K	number of provider types
R_k	the skills set of type k providers
c_k	the fixed cost for hire type k full-time providers
$c_{k,l,t}$	the unitary service cost of type k full-time providers for serve type l customers in day t
$c_{l,t}^u$	the unitary penalty cost due to shortage of service capacity of type l service
$c_{l,t}^o$	the unitary penalty cost due to over-staffing of service capacity of type l service
$\bar{w}$	the upper bound number of full-time providers
$\underline{w}$	the lower bound number of full-time providers
h_k	the service capacity (service hours) of one type k full-time provider in one day
Scenario-dependent Parameters	
$d_{l,t}$	the demand quantity for service type l in day t
$s_{l,t}$	the service time for service type l of provider in day t

random cost $Q(x, y, \xi)$ (third term) subject to uncertainty $\xi \sim \mathbb{P}$. The second stage recourse function minimize the random over-staffing (first-term) and under-staffing (second-term) costs. Constraints (2b) ensure that the total number of hired providers/staff is at least $\underline{w}$ and at most $\bar{w}$ (these can be adjusted by the decision-maker depending on their needs and relevant practical constraints). Constraints (2c) ensure that capacity allocations to service types does not exceed the available capacity. Constraints (3b) compute the over-staffing and under-staffing.

4.1.2. DRO model for EA based on 1-Wasserstein ambiguity set (E-WDHSCP)

In this section, we present our proposed DRO formulation of the HSCP for the EA decision-maker, which does not assume that the probability distributions of demand and service duration are known. Instead, we consider the case where the decision-maker possesses a (possibly small) set $\{\xi^1, \dots, \xi^N\}$ of $\xi := [d, s]^T$. Accordingly, we use this data to construct a 1-Wasserstein ambiguity set (i.e., we use ℓ_1 -norm in the definition of Wasserstein metric), which allow us to derive tractable DRO models, see, e.g., Jiang et al. (2019), Tsang and Shehadeh (2023).

First, let us define the 1-Wasserstein distance. Suppose that random vectors ξ_1 and ξ_2 follow $\mathbb{P}_1$ and $\mathbb{P}_2$, where probability measures are defined on the measurable space $(\mathcal{U}, \mathcal{B})$. The 1-Wasserstein distance $W(\mathbb{P}_1, \mathbb{P}_2)$ between $\mathbb{P}_1$ and $\mathbb{P}_2$ is the minimum transportation cost of moving from $\mathbb{P}_1$ to $\mathbb{P}_2$, where the cost of moving masses ξ_1 to ξ_2 equals to the norm $|\xi_1 - \xi_2|$. Mathematically,

$$W(\mathbb{P}_1, \mathbb{P}_2) := \inf_{\Pi \in \mathcal{P}(\mathbb{P}_1, \mathbb{P}_2)} \left\{ \mathbb{E}_{\Pi} |\xi_1 - \xi_2| \right\}, \quad (4)$$

where $\mathcal{P}(\mathbb{P}_1, \mathbb{P}_2)$ is the set of all joint distributions of (ξ_1, ξ_2) supported on $\mathcal{U}$ with marginals $\mathbb{P}_1$ and $\mathbb{P}_2$. Accordingly, we construct the following 1-Wasserstein ambiguity set:

$$\mathcal{F}(\hat{\mathbb{P}}_{\xi}^N, \epsilon) = \left\{ \mathbb{P}_{\xi} \in \mathcal{P}(\mathcal{U}) \mid W(\mathbb{P}_{\xi}, \hat{\mathbb{P}}_{\xi}^N) \leq \epsilon \right\}, \quad (5)$$

where $\mathcal{P}(\mathcal{U})$ is the set of all probability distributions on $\mathcal{U}$, $\hat{\mathbb{P}}_{\xi}^N = \frac{1}{N} \sum_{n=1}^N \delta_{\xi^n}$ is the empirical distribution of ξ based on the N i.i.d samples with δ being the Dirac measure on ξ , and $\epsilon > 0$ is the radius of the ambiguity set. The set $\mathcal{F}(\hat{\mathbb{P}}_{\xi}^N, \epsilon)$ represents a Wasserstein ball of radius ϵ centered at the empirical distribution $\hat{\mathbb{P}}_{\xi}^N$, where a larger radius ϵ indicates that we seek more robust solutions. Using the ambiguity set $\mathcal{F}(\hat{\mathbb{P}}_{\xi}^N, \epsilon)$, we formulate the E-WDHSCP model as

$$\begin{aligned} (\text{E-WDHSCP}) \quad & \min_{(x, y) \in \mathcal{X}} \left\{ \sum_{k=1}^K c_k x_k + \sum_{k=1}^K \sum_{l=1}^L \sum_{t=1}^T c_{k,l,t} y_{k,l,t} \right. \\ & \left. + \sup_{\mathbb{P}_{\xi} \in \mathcal{F}(\hat{\mathbb{P}}_{\xi}^N, \epsilon)} \mathbb{E}_{\mathbb{P}_{\xi}} [Q(x, y, \xi)] \right\}, \end{aligned} \quad (6)$$

where $\mathcal{X} := \{(x, y) : (2b)-(2d)\}$ is the feasible region of first stage decision variables. In data-driven models such as (6), we often seek asymptotic consistency, i.e., as the sample size N increases, the optimal value of the problem (6) converges to that of problem (2) under the true distribution $\mathbb{P}_{\xi}$, and the optimal solution (x, y) to problem (6) tends to the optimal solution of problem (2). Additionally, if the optimal value of the problem (6) is larger than the optimal value of problem (2) almost surely, then E-WDHSCP provides a safe upper bound guarantee on the expected total cost with any finite data size N . For completeness, in Appendix B.1 – Appendix B.3, we present the adopted proofs for establishing the asymptotic consistency and finite sample guarantee of our E-WDHSCP.

Recall that $Q(\cdot)$ is defined by a minimization problem; hence, in (6), we have an inner max–min problem. As such, it is not straightforward to solve (6) in its presented form. Next, we derive an equivalent formulation of problem (6) that is solvable. First, in Proposition 1, we present an equivalent reformulation of the inner maximization problem $\sup_{\mathbb{P}_{\xi} \in \mathcal{F}(\hat{\mathbb{P}}_{\xi}^N, \epsilon)} \mathbb{E}_{\mathbb{P}_{\xi}} [Q(x, y, \xi)]$ in (6) (see Appendix B.4 for a detailed proof).

Proposition 1. For any $(x, y) \in \mathcal{X}$, problem $\sup_{\mathbb{P}_{\xi} \in \mathcal{F}(\hat{\mathbb{P}}_{\xi}^N, \epsilon)} \mathbb{E}_{\mathbb{P}_{\xi}} [Q(x, y, \xi)]$ in (6) is equivalent to

$$\inf_{\gamma} \left\{ \epsilon \gamma + \frac{1}{N} \sum_{n=1}^N \sup_{\xi \in \mathcal{U}} \left\{ Q(x, y, \xi) - \gamma |\xi - \xi^n| \right\} \right\}, \quad (7a)$$

$$\text{s.t. } \gamma \geq 0. \quad (7b)$$

Again, problem (7) involves an inner max–min problem that is not straightforward to solve in its presented form. However, as shown in the poof of Proposition 2 in Appendix B.5, we leverage the structural properties of the recourse $Q(\cdot)$ to derive an equivalent linear programming (LP) reformulation of the inner problem $\sup_{\xi \in \mathcal{U}} \{Q(x, y, \xi) - \gamma |\xi - \xi^n|\}$.

Proposition 2. For fixed x, y, γ , and ξ^n , problem $\max_{\xi \in \mathcal{U}} \{Q(x, y, \xi) - \gamma |\xi - \xi^n|\}$ is equivalent to the following LP.

$$\min \sum_{l=1}^L \sum_{t=1}^T \eta_{l,t}^n \quad (8a)$$

$$\begin{aligned} \text{s.t. } \eta_{l,t}^n & \geq -d_{l,t} s_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o \\ & - \gamma (\hat{d}_{l,t}^n - d_{l,t}) - \gamma (\hat{s}_{l,t}^n - s_{l,t}), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8b)$$

$$\eta_{l,t}^n \geq -\bar{d}_{l,t} s_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o$$

$$-\gamma(\bar{d}_{l,t} - \hat{d}_{l,t}^n) - \gamma(\hat{s}_{l,t}^n - \underline{s}_{l,t}), \quad \forall l \in [L], t \in [T], \quad (8c)$$

$$\eta_{l,t}^n \geq -\hat{d}_{l,t}^n \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \gamma(\hat{s}_{l,t}^n - \underline{s}_{l,t}), \quad \forall l \in [L], t \in [T], \quad (8d)$$

$$\begin{aligned} \eta_{l,t}^n &\geq -\underline{d}_{l,t} \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o \\ &\quad - \gamma(\hat{d}_{l,t}^n - \underline{d}_{l,t}) - \gamma(\bar{s}_{l,t} - \hat{s}_{l,t}^n), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8e)$$

$$\begin{aligned} \eta_{l,t}^n &\geq -\bar{d}_{l,t} \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o \\ &\quad - \gamma(\bar{d}_{l,t} - \hat{d}_{l,t}^n) - \gamma(\bar{s}_{l,t} - \hat{s}_{l,t}^n), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8f)$$

$$\eta_{l,t}^n \geq -\hat{d}_{l,t}^n \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \gamma(\bar{s}_{l,t} - \hat{s}_{l,t}^n), \quad \forall l \in [L], t \in [T], \quad (8g)$$

$$\eta_{l,t}^n \geq -\underline{d}_{l,t} \hat{s}_{l,t}^n c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \gamma(\hat{d}_{l,t}^n - \underline{d}_{l,t}), \quad \forall l \in [L], t \in [T], \quad (8h)$$

$$\eta_{l,t}^n \geq -\bar{d}_{l,t} \hat{s}_{l,t}^n c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \gamma(\bar{d}_{l,t} - \hat{d}_{l,t}^n), \quad \forall l \in [L], t \in [T], \quad (8i)$$

$$\eta_{l,t}^n \geq -\hat{d}_{l,t}^n \hat{s}_{l,t}^n c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o, \quad \forall l \in [L], t \in [T], \quad (8j)$$

$$\begin{aligned} \eta_{l,t}^n &\geq \underline{d}_{l,t} \underline{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u \\ &\quad - \gamma(\hat{d}_{l,t}^n - \underline{d}_{l,t}) - \gamma(\hat{s}_{l,t}^n - \underline{s}_{l,t}), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8k)$$

$$\begin{aligned} \eta_{l,t}^n &\geq \bar{d}_{l,t} \underline{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u \\ &\quad - \gamma(\bar{d}_{l,t} - \hat{d}_{l,t}^n) - \gamma(\hat{s}_{l,t}^n - \underline{s}_{l,t}), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8l)$$

$$\eta_{l,t}^n \geq \hat{d}_{l,t}^n \underline{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \gamma(\hat{s}_{l,t}^n - \underline{s}_{l,t}), \quad \forall l \in [L], t \in [T], \quad (8m)$$

$$\begin{aligned} \eta_{l,t}^n &\geq \underline{d}_{l,t} \bar{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u \\ &\quad - \gamma(\hat{d}_{l,t}^n - \underline{d}_{l,t}) - \gamma(\bar{s}_{l,t} - \hat{s}_{l,t}^n), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8n)$$

$$\begin{aligned} \eta_{l,t}^n &\geq \bar{d}_{l,t} \bar{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u \\ &\quad - \gamma(\bar{d}_{l,t} - \hat{d}_{l,t}^n) - \gamma(\bar{s}_{l,t} - \hat{s}_{l,t}^n), \quad \forall l \in [L], t \in [T], \end{aligned} \quad (8o)$$

$$\eta_{l,t}^n \geq \hat{d}_{l,t}^n \bar{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \gamma(\bar{s}_{l,t} - \hat{s}_{l,t}^n), \quad \forall l \in [L], t \in [T], \quad (8p)$$

$$\eta_{l,t}^n \geq \underline{d}_{l,t} \hat{s}_{l,t}^n c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \gamma(\hat{d}_{l,t}^n - \underline{d}_{l,t}), \quad \forall l \in [L], t \in [T], \quad (8q)$$

$$\eta_{l,t}^n \geq \bar{d}_{l,t} \hat{s}_{l,t}^n c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \gamma(\bar{d}_{l,t} - \hat{d}_{l,t}^n), \quad \forall l \in [L], t \in [T], \quad (8r)$$

$$\eta_{l,t}^n \geq \hat{d}_{l,t}^n \hat{s}_{l,t}^n c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u, \quad \forall l \in [L], t \in [T]. \quad (8s)$$

Combining the inner problem in the form of (8) with the outer minimization problems in (6) and (7), we derive the following equivalent MILP reformulation of the E-WDHSCP model.

$$(E\text{-}WDHSCP) \min_{\mathbf{x}, \mathbf{y}, \gamma, \eta} \left\{ \sum_{k=1}^K c_k x_k + \sum_{k=1}^K \sum_{l=1}^L \sum_{t=1}^T c_{k,l,t} y_{k,l,t} + \epsilon \gamma + \frac{1}{N} \sum_{n=1}^N \sum_{l=1}^L \sum_{t=1}^T \eta_{l,t}^n \right\} \quad (9a)$$

$$\text{s.t. } (\mathbf{x}, \mathbf{y}) \in \mathcal{X}, \gamma \geq 0, \eta^n \in \{(8b)\text{--}(8s)\}_n, \forall n \in [N]. \quad (9b)$$

4.1.3. DRO model for EA based on mean-support ambiguity set (E-MDHSCP)

In this section, we present our proposed DRO formulation with mean-support ambiguity set for the EA decision-maker (denoted as E-MDHSCP). We define the following notation for this model. We let $\mathbb{E}_{\mathbb{P}}$ as the expectation under $\mathbb{P}$. Finally, we denote $\boldsymbol{\mu} := \mathbb{E}_{\mathbb{P}}[\xi] = [\boldsymbol{\mu}^d, \boldsymbol{\mu}^s]^\top$. Using this notation, we construct the following mean-support ambiguity set:

$$\mathcal{F}(\mathcal{U}, \boldsymbol{\mu}) := \left\{ \mathbb{P} \in \mathcal{P}(\mathcal{U}) \mid \mathbb{E}_{\mathbb{P}}[\xi] = \boldsymbol{\mu}, \mathbb{P}(\mathcal{U}) = 1 \right\}. \quad (10)$$

Using the ambiguity set $\mathcal{F}(\mathcal{U}, \boldsymbol{\mu})$, we formulate our DRO model of the HSCP with EA decision-maker as follow:

$$(E\text{-}MDHSCP) \min_{(\mathbf{x}, \mathbf{y}) \in \mathcal{X}'} \left\{ \sum_{k=1}^K c_k x_k + \sum_{k=1}^K \sum_{l=1}^L \sum_{t=1}^T c_{k,l,t} y_{k,l,t} + \sup_{\mathbb{P} \in \mathcal{F}(\mathcal{U}, \boldsymbol{\mu})} \mathbb{E}_{\mathbb{P}}[Q(\mathbf{x}, \mathbf{y}, \xi)] \right\}. \quad (11)$$

Formulation (11) seeks first stage decisions $\mathbf{x}$ and $\mathbf{y}$ that minimizes the summation of fixed hiring cost (first term), the capacity allocation cost (second term) and the worst-case (maximum) expectations of the second-stage operational cost (third term), where the expectation is taken over all distributions residing in $\mathcal{F}(\mathcal{U}, \boldsymbol{\mu})$. Note that this model is also not directly solvable in the presented form. In Appendix B.6, we derive an equivalent MILP reformulation. Herein, we present this MILP in (12).

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \alpha, \beta, \eta} \left\{ \sum_{k=1}^K c_k x_k + \sum_{k=1}^K \sum_{l=1}^L \sum_{t=1}^T c_{k,l,t} y_{k,l,t} \right. \\ \left. + \sum_{l=1}^L \sum_{t=1}^T \mu_{l,t}^d \alpha_{l,t} + \sum_{l=1}^L \sum_{t=1}^T \mu_{l,t}^s \beta_{l,t} + \sum_{l=1}^L \sum_{t=1}^T \eta_{l,t} \right\} \end{aligned} \quad (12a)$$

$$\text{s.t. } (\mathbf{x}, \mathbf{y}) \in \mathcal{X}, \quad (12b)$$

$$\eta_{l,t} \geq \bar{d}_{l,t} \bar{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \bar{d}_{l,t} \alpha_{l,t} - \bar{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12c)$$

$$\eta_{l,t} \geq -\underline{d}_{l,t} \underline{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \underline{d}_{l,t} \alpha_{l,t} - \underline{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12d)$$

$$\eta_{l,t} \geq -\bar{d}_{l,t} \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \bar{d}_{l,t} \alpha_{l,t} - \bar{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12e)$$

$$\eta_{l,t} \geq \underline{d}_{l,t} \bar{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \underline{d}_{l,t} \alpha_{l,t} - \bar{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12f)$$

$$\eta_{l,t} \geq \bar{d}_{l,t} \underline{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \bar{d}_{l,t} \alpha_{l,t} - \underline{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12g)$$

$$\eta_{l,t} \geq \underline{d}_{l,t} \underline{s}_{l,t} c_{l,t}^u - \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^u - \underline{d}_{l,t} \alpha_{l,t} - \underline{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12h)$$

$$\eta_{l,t} \geq -\bar{d}_{l,t} \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \bar{d}_{l,t} \alpha_{l,t} - \bar{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T], \quad (12i)$$

$$\eta_{l,t} \geq -\underline{d}_{l,t} \bar{s}_{l,t} c_{l,t}^o + \sum_{k:l \in R_k} y_{k,l,t} c_{l,t}^o - \underline{d}_{l,t} \alpha_{l,t} - \bar{s}_{l,t} \beta_{l,t}, \quad \forall l \in [L], t \in [T]. \quad (12j)$$

4.2. Models for the FA decision-maker

4.2.1. Two-stage SP for FA

In this section, we present our proposed SP formulation of the HSCP problem considering the FA decision-maker, denoted as F-SP. In this model, we keep variables x_k , $\forall k \in [K]$ as first stage decisions, and move $y_{k,l,t}$ to the second-stage, for all $k \in [K], l \in [L], t \in [T]$. In addition, we define non-negative continuous variables $o_{k,t}$ representing the capacity surplus of type $k \in [K]$ provider in day $t \in [T]$. This is to account for scenarios where the total capacity of hired providers is larger than the total realized demand. Accordingly, our F-SP model can now be stated as follows:

$$(F-SP) \min_x \sum_{k=1}^K c_k x_k + \mathbb{E}[Q^A(\mathbf{x}, \xi)] \quad (13a)$$

$$\text{s.t. } \underline{w} \leq \sum_{k=1}^K x_k \leq \bar{w}, \quad (13b)$$

$$x_k \in \mathbb{Z}_+, \quad \forall k \in [K], \quad (13c)$$

where for each feasible $\mathbf{x}$ and a realization of ξ ,

$$Q^A(\mathbf{x}, \xi) = \min_{y, o, u} \sum_{k=1}^K \sum_{l=1}^L \sum_{t=1}^T c_{k,l,t} y_{k,l,t} + \sum_{k=1}^K \sum_{t=1}^T c_{k,t}^o o_{k,t} + \sum_{l=1}^L \sum_{t=1}^T c_{l,t}^u u_{l,t} \quad (14a)$$

$$\text{s.t. } \sum_{l \in R_k} y_{k,l,t} + o_{k,t} = x_k h_k, \quad \forall k \in [K], t \in [T], \quad (14b)$$

$$\sum_{k:l \in R_k} y_{k,l,t} + u_{l,t} = d_{l,t} s_{l,t}, \quad \forall l \in [L], t \in [T], \quad (14c)$$

$$y_{k,l,t}, o_{k,t}, u_{l,t} \in \mathbb{R}_+, \quad \forall k \in [K], l \in [L], t \in [T]. \quad (14d)$$

Formulation (13) seeks to find staffing decisions ($\mathbf{x}$) that minimize the sum of (1) fixed employment cost (first term), and (2) expectation of the second stage (recourse) random cost $Q^A(\mathbf{x}, \xi)$ (second term) subject to uncertainty $\xi \sim \mathbb{P}$. The second stage recourse function minimize the random assignment cost (first-term), over-staffing (second-term) and under-staffing (third term) costs. Constraints (14b) compute the over-staffing. Constraints (14c) compute the under-staffing.

4.2.2. DRO model for FA based on 1-Wasserstein ambiguity set (F-WDHSCP)

In this section, we present our proposed DRO formulation of the HSCP with FA decision-maker, denoted as F-WDHSCP. Using the ambiguity set $\mathcal{F}(\hat{\mathbb{P}}_\xi^N, \epsilon)$ defined in (5), we formulate our DRO model of the HSCP with FA decision-maker as follow:

$$(F-WDHSCP) \min_{\mathbf{x} \in \mathcal{X}_2} \left\{ \sum_{k=1}^K c_k x_k + \sup_{\mathbb{P}_\xi \in \mathcal{F}(\hat{\mathbb{P}}_\xi^N, \epsilon)} \mathbb{E}_{\mathbb{P}_\xi}[Q^A(\mathbf{x}, \xi)] \right\}, \quad (15)$$

where $\mathcal{X}_2 := \{(13b)-(13c)\}$ is the feasible region of first stage decision variables. Formulation (15) seeks the first stage decisions $\mathbf{x}$ that minimize the sum of fixed employment cost (first term) and the worst-case (maximum) expectations of the second-stage operational cost (second term), where the expectation is taken over all distributions residing in $\mathcal{F}(\hat{\mathbb{P}}_\xi^N, \epsilon)$. In Appendix C.1, we derive the following equivalent solvable reformulation of this model.

$$(F-WDHSCP) \min_{\mathbf{x} \in \mathcal{X}_2, \gamma} \left\{ \sum_{k=1}^K c_k x_k + \epsilon \gamma + \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T \delta_t^n \right\} \quad (16a)$$

$$\text{s.t. } \delta_t^n \geq f_t^n(\mathbf{x}, \gamma), \quad t \in [T], n \in [N], \quad (16b)$$

where $f_t^n(\mathbf{x}, \gamma)$ is as follows:

$$\begin{aligned} \max \quad & \sum_{k=1}^K x_k h_k \lambda_{k,t}^n + \sum_{l=1}^L \left\{ \hat{d}_{l,t}^n \hat{s}_{l,t}^n \rho_{l,t}^n - \hat{d}_{l,t}^n \Delta s_{l,t}^n \phi_{l,t}^n \right. \\ & + \hat{d}_{l,t}^n \Delta \bar{s}_{l,t}^n \zeta_{l,t}^n - \hat{s}_{l,t}^n \Delta \underline{d}_{l,t}^n \phi_{l,t}^n \\ & + \Delta \underline{d}_{l,t}^n \Delta s_{l,t}^n \chi_{l,t}^n - \Delta \underline{d}_{l,t}^n \Delta \bar{s}_{l,t}^n \omega_{l,t}^n + \hat{s}_{l,t}^n \Delta \bar{d}_{l,t}^n \phi_{l,t}^n \\ & \left. - \Delta \bar{d}_{l,t}^n \Delta s_{l,t}^n \pi_{l,t}^n + \Delta \bar{d}_{l,t}^n \Delta \bar{s}_{l,t}^n \delta_{l,t}^n \right\} \\ & - \gamma \sum_{l=1}^L (z_{l,t}^n \Delta \bar{d}_{l,t}^n + g_{l,t}^n \Delta \underline{d}_{l,t}^n + \beta_{l,t}^n \Delta \bar{s}_{l,t}^n + \alpha_{l,t}^n \Delta \underline{s}_{l,t}^n) \end{aligned} \quad (17a)$$

$$\text{s.t. } \rho_{l,t} + \lambda_{k,t} \leq c_{k,l,t}, \quad \forall k \in [K], l \in R_k, \quad (17b)$$

$$\rho_{l,t} \leq c_{l,t}^u, \quad \forall l \in [L], \quad (17c)$$

$$\lambda_{k,t} \leq c_{k,t}^o, \quad \forall k \in [K], \quad (17d)$$

$$g_{l,t}^n + z_{l,t}^n \leq 1, \quad \alpha_{l,t}^n + \beta_{l,t}^n \leq 1, \quad \forall l \in [L], \quad (17e)$$

$$\zeta_{l,t}^n - g_{l,t}^n \leq 0, \quad \zeta_{l,t}^n - \alpha_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17f)$$

$$\zeta_{l,t}^n \geq 0, \quad \zeta_{l,t}^n - g_{l,t}^n - \alpha_{l,t}^n \geq -1, \quad \forall l \in [L], \quad (17g)$$

$$\kappa_{l,t}^n - g_{l,t}^n \leq 0, \quad \kappa_{l,t}^n - \beta_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17h)$$

$$\kappa_{l,t}^n \geq 0, \quad \kappa_{l,t}^n - g_{l,t}^n - \beta_{l,t}^n \geq -1, \quad \forall l \in [L], \quad (17i)$$

$$v_{l,t}^n - z_{l,t}^n \leq 0, \quad v_{l,t}^n - \alpha_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17j)$$

$$v_{l,t}^n \geq 0, \quad v_{l,t}^n - z_{l,t}^n - \alpha_{l,t}^n \geq -1, \quad \forall l \in [L], \quad (17k)$$

$$i_{l,t}^n - z_{l,t}^n \leq 0, \quad i_{l,t}^n - \beta_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17l)$$

$$i_{l,t}^n \geq 0, \quad i_{l,t}^n - z_{l,t}^n - \beta_{l,t}^n \geq -1, \quad \forall l \in [L], \quad (17m)$$

$$\phi_{l,t}^n - \rho_{l,t}^n \alpha_{l,t}^n \geq 0, \quad \phi_{l,t}^n - c_{l,t}^u \alpha_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17n)$$

$$\phi_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^n (1 - \alpha_{l,t}^n) \leq 0,$$

$$\phi_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u (1 - \alpha_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17o)$$

$$\theta_{l,t}^n - \rho_{l,t}^n \beta_{l,t}^n \geq 0, \quad \theta_{l,t}^n - c_{l,t}^u \beta_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17p)$$

$$\theta_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^n (1 - \beta_{l,t}^n) \leq 0,$$

$$\theta_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u (1 - \beta_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17q)$$

$$\phi_{l,t}^n - \rho_{l,t}^n g_{l,t}^n \geq 0, \quad \phi_{l,t}^n - c_{l,t}^u g_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17r)$$

$$\phi_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^n (1 - g_{l,t}^n) \leq 0,$$

$$\phi_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u (1 - g_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17s)$$

$$\chi_{l,t}^n - \rho_{l,t}^n \zeta_{l,t}^n \geq 0, \quad \chi_{l,t}^n - c_{l,t}^u \zeta_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17t)$$

$$\chi_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^n (1 - \zeta_{l,t}^n) \leq 0,$$

$$\kappa_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u(1 - \zeta_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17u)$$

$$\varpi_{l,t}^n - \rho_{l,t}^n \kappa_{l,t}^n \geq 0, \quad \varpi_{l,t}^n - c_{l,t}^u \kappa_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17v)$$

$$\varpi_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^u(1 - \kappa_{l,t}^n) \leq 0, \quad \varpi_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u(1 - \kappa_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17w)$$

$$\zeta_{l,t}^n - \rho_{l,t}^n \kappa_{l,t}^n \geq 0, \quad \zeta_{l,t}^n - c_{l,t}^u \kappa_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17x)$$

$$\zeta_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^u(1 - \kappa_{l,t}^n) \leq 0, \quad \zeta_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u(1 - \kappa_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17y)$$

$$\pi_{l,t}^n - \rho_{l,t}^n v_{l,t}^n \geq 0, \quad \pi_{l,t}^n - c_{l,t}^u v_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17z)$$

$$\pi_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^u(1 - v_{l,t}^n) \leq 0, \quad \pi_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u(1 - v_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17aa)$$

$$\delta_{l,t}^n - \rho_{l,t}^n i_{l,t}^n \geq 0, \quad \delta_{l,t}^n - c_{l,t}^u i_{l,t}^n \leq 0, \quad \forall l \in [L], \quad (17ab)$$

$$\delta_{l,t}^n - \rho_{l,t}^n + \rho_{l,t}^u(1 - i_{l,t}^n) \leq 0, \quad \delta_{l,t}^n - \rho_{l,t}^n + c_{l,t}^u(1 - i_{l,t}^n) \geq 0, \quad \forall l \in [L], \quad (17ac)$$

$$g_{l,t}^n, z_{l,t}^n, \alpha_{l,t}^n, \beta_{l,t}^n \in \{0, 1\}, \quad \forall l \in [L]. \quad (17ad)$$

where $\Delta \bar{d}_{l,t}^n = \bar{d}_{l,t}^n - \hat{d}_{l,t}^n$, $\Delta \underline{d}_{l,t}^n = \hat{d}_{l,t}^n - \underline{d}_{l,t}^n$, $\Delta \bar{s}_{l,t}^n = \bar{s}_{l,t}^n - \hat{s}_{l,t}^n$, and $\Delta \underline{s}_{l,t}^n = \hat{s}_{l,t}^n - \underline{s}_{l,t}^n$, for all $l \in [L]$, $t \in [T]$ and $n \in [N]$.

Note that McCormick inequalities (17n)–(17ad) in formulation (17) involve big-M coefficients $\rho_{l,t}^u$, for all $l \in [L]$ and $t \in [T]$, that can undermine computational efficiency if they are set too large. Therefore, in Proposition 3, we derive tight lower bounds (i.e., $\rho_{l,t}^u$) on variables $\rho_{l,t}$ to strengthen the MILP reformulation (see Appendix C.2 for a detailed proof).

Proposition 3. $\rho_{l,t}^u = \min_{k: l \in R_k} (c_{k,l,t} - c_{k,t}^o)$ is a valid lower bound on $\rho_{l,t}$, for all $l \in [L]$ and $t \in [T]$.

4.2.3. DRO model for FA based on mean-support ambiguity set (F-MDHS CP)

In this section, using the ambiguity set $\mathcal{F}(\mathcal{U}, \mu)$ defined in (10), we formulate the following DRO model based on mean-support ambiguity set for the FA decision-maker (denoted as F-MDHS CP).

$$(\text{F-MDHS CP}) \quad \min_{\mathbf{x} \in \mathcal{X}_2} \left\{ \sum_{k=1}^K c_k x_k + \sup_{\mathbb{P} \in \mathcal{F}(\mathcal{U}, \mu)} \mathbb{E}_{\mathbb{P}}[Q^A(\mathbf{x}, \xi)] \right\}, \quad (18)$$

Formulation (18) seeks the first stage decisions $\mathbf{x}$ that minimize the sum of fixed employment cost (first term) and the worst-case (maximum) expectations of the second-stage operational cost (second term), where the expectation is taken over all distributions residing in $\mathcal{F}(\mathcal{U}, \mu)$. In Appendix C.1, we derive the following equivalent solvable reformulation of this model.

$$\min_{\mathbf{x} \in \mathcal{X}_2, \alpha, \beta} \left\{ \sum_{k=1}^K c_k x_k + \sum_{t=1}^T \sum_{l=1}^L (\mu_{l,t}^d \alpha_{l,t} + \mu_{l,t}^s \beta_{l,t}) + \sum_{t=1}^T \delta_t \right\} \quad (19a)$$

$$\text{s.t.} \quad \delta_t \geq f_t(\mathbf{x}, \alpha, \beta), \quad t \in [T], \quad (19b)$$

where $f_t(\mathbf{x}, \alpha, \beta)$ in (19b) is as follow:

$$f_t(\mathbf{x}, \alpha, \beta) := \max_{\substack{\rho, \lambda, g, z, \\ v, r, q, e}} \left\{ \sum_{k=1}^K x_k h_k \lambda_{k,t} - \sum_{l=1}^L [\underline{d}_{l,t} \alpha_{l,t} \right.$$

$$+ \Delta d_{l,t} g_{l,t} \alpha_{l,t} + \underline{s}_{l,t} \beta_{l,t} + \Delta s_{l,t} z_{l,t} \beta_{l,t}] \\ + \sum_{l=1}^L \left[\underline{d}_{l,t} \underline{s}_{l,t} \rho_{l,t} + \Delta s_{l,t} \underline{d}_{l,t} b_{l,t} \right. \\ \left. + \Delta d_{l,t} \underline{s}_{l,t} q_{l,t} + \Delta d_{l,t} \Delta s_{l,t} e_{l,t} \right] \Big\} \quad (20a)$$

$$\text{s.t.} \quad (17b) \text{--}(17d), \quad (20b)$$

$$\rho_{l,t} \geq \rho_{l,t}^u, \quad v_{l,t} - g_{l,t} \leq 0, \quad \forall l \in [L], \quad (20c)$$

$$v_{l,t} \geq 0, \quad v_{l,t} - z_{l,t} \leq 0, \quad \forall l \in [L], \quad (20d)$$

$$v_{l,t} - g_{l,t} - z_{l,t} \geq -1, \quad \forall l \in [L], \quad (20e)$$

$$e_{l,t} - \rho_{l,t} v_{l,t} \geq 0, \quad e_{l,t} - c_{l,t}^u v_{l,t} \leq 0, \quad \forall l \in [L], \quad (20f)$$

$$e_{l,t} - \rho_{l,t} + \rho_{l,t}^u(1 - v_{l,t}) \leq 0, \quad e_{l,t} - \rho_{l,t} + c_{l,t}^u(1 - v_{l,t}) \geq 0, \quad \forall l \in [L], \quad (20g)$$

$$q_{l,t} - \rho_{l,t} g_{l,t} \geq 0, \quad q_{l,t} - c_{l,t}^u g_{l,t} \leq 0, \quad \forall l \in [L], \quad (20h)$$

$$q_{l,t} - \rho_{l,t} + \rho_{l,t}^u(1 - g_{l,t}) \leq 0, \quad q_{l,t} - \rho_{l,t} + c_{l,t}^u(1 - g_{l,t}) \geq 0, \quad \forall l \in [L], \quad (20i)$$

$$b_{l,t} - \rho_{l,t} v_{l,t} \geq 0, \quad b_{l,t} - c_{l,t}^u v_{l,t} \leq 0, \quad \forall l \in [L], \quad (20j)$$

$$b_{l,t} - \rho_{l,t} + \rho_{l,t}^u(1 - v_{l,t}) \leq 0, \quad b_{l,t} - \rho_{l,t} + c_{l,t}^u(1 - v_{l,t}) \geq 0, \quad \forall l \in [L], \quad (20k)$$

$$g_{l,t}, z_{l,t} \in \{0, 1\}, v_{l,t}, b_{l,t}, q_{l,t}, e_{l,t}, \rho_{l,t} \in \mathbb{R} \quad \forall l \in [L]. \quad (20l)$$

where $\Delta d_{l,t} = \bar{d}_{l,t} - \underline{d}_{l,t}$ and $\Delta s_{l,t} = \bar{s}_{l,t} - \underline{s}_{l,t}$, for all $l \in [L]$ and $t \in [T]$.

5. Solution approaches

Note that the E-MDHS CP and E-WDHS CP model can be implemented and solved directly using off-the-shelf optimization software. In contrast, the F-MDHS CP and F-WDHS CP models involve maximization problems in constraints (19b) and (16b). Thus, these formulations are not directly solvable. In this section, we present a column-and-constraint generation (C&CG) algorithm to solve the F-MDHS CP models. We present a C&CG algorithm for F-WDHS CP model in Appendix D.1.

Algorithm 1 presents our C&CG algorithm for the F-MDHS CP model. The master problem 21 in Algorithm 1 employs scenario-based constraints (21d)–(21f) and the sub-problem generates demand and service time scenarios and optimality cuts from the worst-case distribution. The algorithm proceeds as follows. At each iteration, we first solve the

master problem, which includes only a subset of demand and service time scenarios, and obtain the optimal solution (i.e., hiring decisions). Thus, by considering a subset of demand and service time scenarios, the master problem is a relaxation of the original problem and thus provides a lower bound on the optimal value of F-MDHSCP. Then, given the optimal solution from the master problem, we identify demand d^* and service time s^* scenarios by solving the sub-problem. Note that solutions of the master problem are feasible to the original problem, then by solving the sub-problem using these solutions, we obtain an upper bound. Third, we pass the identified demand and service time scenarios to the master problem on an as-needed basis, along with introducing relevant second-stage variables and constraints. Then, we solve the master problem again with the new information (a larger set of scenarios $\mathcal{O}$) from the sub-problems. This process continues until the gap between the lower and upper bound on the optimal value of the problem obtained in each iteration satisfies a predetermined termination tolerance.

Recall that the F-MDHSCP formulation's recourse problem is feasible for every feasible first-stage decision and $(d, s) \in \mathcal{U}$. Thus, we do not need to add any feasibility cuts. Moreover, we only have a finite number of scenarios. Thus, the algorithm terminates in a finite number of iterations.

5.1. Valid inequalities

In this section, we derive the following valid inequalities to strengthen the master problem of the C&CG and thus improve convergence.

(1) Valid lower bound (LB) inequalities

Observe that the Dirac measure on μ lies in $F(\mathcal{U}, \mu)$. Therefore, we have

$$\sup_{\mathbb{P} \in F(\mathcal{U}, \mu)} \mathbb{E}_{\mathbb{P}}[Q^A(x, \xi)] \geq Q^A(x, \mu) =: L, \quad (23)$$

where the lower bound is a deterministic problem with a single scenario $\xi = \mu$. This means we can impose the following constraint

$$\delta_l + \sum_{i=1}^L (d_{l,i} \alpha_{l,i} + s_{l,i} \beta_{l,i}) \geq L, \quad (24)$$

which serves as a global lower bound on the objective of the DRO model. Second, from (23), for any given first-stage decision, the recourse value with $\xi = \mu$ provides a lower bound. Therefore, in the initialization step of the C&CG method, we could include the scenario $\xi = \mu$.

(2) Valid upper and lower bound of variables α and β

In Proposition 4, we derive valid lower $(\underline{\alpha}, \underline{\beta})$ and upper $(\bar{\alpha}, \bar{\beta})$ bounds on variables α and β in the master problem 21 of the C&CG algorithm (see Appendix D.2 for a proof). In Section 6.4, we demonstrate the computational advantages gained by incorporating these valid inequalities into the master problem 21 of our proposed C&CG algorithm.

Proposition 4. *The following bounds are valid lower and upper bounds on variables α and β*

$$\underline{\alpha}_{l,t} \leq \alpha_{l,t} \leq \bar{\alpha}_{l,t}, \quad \forall l \in [L], t \in [T], \quad (25a)$$

$$\underline{\beta}_{l,t} \leq \beta_{l,t} \leq \bar{\beta}_{l,t}, \quad \forall l \in [L], t \in [T], \quad (25b)$$

where $\bar{\alpha}_{l,t} = \bar{s}_{l,t} c_{l,t}^u$, $\underline{\alpha}_{l,t} = -\bar{s}_{l,t} |\rho_{l,t}|$, $\bar{\beta}_{l,t} = \bar{d}_{l,t} c_{l,t}^u$, and $\underline{\beta}_{l,t} = -\bar{d}_{l,t} |\rho_{l,t}|$.

6. Computational experiments

In this section, we compare the computational and operational performance of the proposed models for each decision maker and derive insights into the HSCP problem. In Section 6.1, we describe the set of test instances that we constructed and discuss other experimental

setups. In Section 6.2, we obtain near-optimal solutions to the E-SP and F-SP via their sample average approximations (SAA). In Section 6.3, we compare solution times of the proposed models. In Section 6.4, we demonstrate efficiency of the proposed valid inequalities for the F-MDHSCP model. In Section 6.5, we compare optimal solutions to the SP and DRO models for the EA decision maker. In Section 6.6, we compare the optimal solutions to the SP and DRO models for the FA decision maker. Finally, we compare the solutions of EA and FA models in Section 6.7.

6.1. Description of experiments

To our knowledge, there are no standard benchmark instances in the literature for the HSCP problem. Therefore, to test the computational efficacy of our proposed models and approaches we construct 15 HSCP instances based on the same parameters settings and assumptions made in recent related literature (see e.g. Lanzarone and Matta, 2014; Rodriguez et al., 2015; Zhan and Wan, 2018; Zhan et al., 2021; Zheng et al., 2021). Each of these instances is characterized by the number of service types ($L \in \{4, 6\}$) as in Mankowska et al., 2014; Rodriguez et al., 2015; Rodriguez-Verjan et al., 2018; Zheng et al., 2021), provider types ($K \in \{4, 6, 8\}$), days in the planning horizon $T \in \{30, 90, 180\}$. We summarize these instances in Table 2. In addition, we construct an instance (denoted as real case; see details below) based on the data from a home services provider in Beijing (Pinetree Care). Pinetree provides two types of home services (i.e., rehabilitation nursing and health status assessment). They provided us with demand information for each type of service (analyzed earlier in Section 2). Table 3 summarizes the statistics of the daily demand for each type of service.

We set the provider standard work hours h_k to 480 min (i.e., 8 h, see, e.g., Zhan and Wan, 2018). We generate the monthly fixed cost for employing one full-time provider, c_k , from $U[3000, 6000]$ as in Rodriguez-Verjan et al. (2018). We generate the quarterly and semi-annual cost, c_k , from $U[9000, 18000]$ and $U[18000, 36000]$, respectively. Given that multi-skills (cross-trained) providers are more costly, we generated their additional monthly cost from $U[500, 1000]$. To ensure that the FA models will allocate providers capacity to different service types, we have $c_{l,t}^u + c_{k,t}^o > c_{k,l,t}$ for all k, l and t . This is because it is always optimal to set $y_{k,l,t} = 0$ when $c_{l,t}^u + c_{k,t}^o < c_{k,l,t}$.

We followed the same procedures in the literature to generate random parameters as follows. For each of the 15 HSCP instances in Table 2, we generate the mean values of the demand, $\mu_{l,t}^d$, from $U[40; 60]$ and set the standard deviation $\sigma_{l,t}^d = \gamma_{l,t}^d \mu_{l,t}^d$, where $\gamma_{l,t}^d$ is generated from $U[0.5, 1]$, for each $l \in [L]$ and $t \in [T]$. For the real case instance, we set the demand's mean, standard deviation, and support to their actual values reported in Table 3. Because we do not have data on service time, we generate the mean service time $\mu_{l,t}^s$ from $U[40; 60]$ and set $\sigma_{l,t}^s = \gamma_{l,t}^s \mu_{l,t}^s$, where $\gamma_{l,t}^s$ is generated from $U[0.5, 1]$, for each $l \in [L]$ and $t \in [T]$.

For each of the 15 HSCP instances, we sample N scenarios of $d_{1,1}^n, \dots, d_{L,T}^n$, for $n = 1, \dots, N$, by following lognormal (LogN) distributions with the generated μ^d and σ^d . We truncated the generated demand scenarios for each HSCP instance on demand range $[\underline{d}, \bar{d}] = [40, 60]$ (as in Rodriguez et al., 2015; Rodriguez-Verjan et al., 2018) and for the real case instance on the demand range reported in Table 3. Similarly, for each instance, we generated N scenarios of $s_{1,1}^n, \dots, s_{L,T}^n$, for $n = 1, \dots, N$, by following LogN distributions with the generated μ^s and σ^s . We truncated these scenarios on $[\underline{s}, \bar{s}] = [20, 80]$. We round each generated parameter to the nearest integer. We chose the LogN to generate the in-sample data because it is typically used to model service time and demand in the home service, scheduling, and other literature (Cayirli et al., 2006; Jiang et al., 2019; Tsang and Shehadeh, 2023).

For DRO models based on 1-Wasserstein ambiguity set, the Wasserstein ball's radius ϵ is an input parameter. Note that a larger ϵ implies that we seek a more distributionally robust solution. In our study,

Algorithm 1: Column-and-constraint generation for F-MDHSCP.**Step 1. Initialization.** Set $LB = 0$, $UB = +\infty$, $R = 0$, $\mathbf{O} = \emptyset$ **Step 2. Master Problem.** Solve the following master problem:

$$Z = \min_{\mathbf{x}, \alpha, \beta, \delta} \left\{ \sum_{k=1}^K c_k x_k + \sum_{t=1}^T \sum_{l=1}^L (\mu_{l,t}^d \alpha_{l,t} + \mu_{l,t}^s \beta_{l,t}) + \sum_{t=1}^T \delta_t \right\} \quad (21a)$$

$$\text{s.t. } \mathbf{x} \in \mathcal{X}_2, \quad (21b)$$

$$\delta_t \geq \sum_{k=1}^K \sum_{l=1}^L c_{k,l,t} y_{k,l,t}^r + \sum_{k=1}^K c_{k,t}^o o_{k,t}^r + \sum_{l=1}^L c_{l,t}^u u_{l,t}^r - \sum_{l=1}^L (d_{l,t}^r \alpha_{l,t} + s_{l,t}^r \beta_{l,t}), \forall t \in [T], \forall r \in \mathbf{O}, \quad (21c)$$

$$\sum_{l \in R_k} y_{k,l,t}^r + o_{k,t}^r = x_k h_k, \quad \forall k \in [K], t \in [T], r \leq R, \quad (21d)$$

$$\sum_{k: l \in R_k} y_{k,l,t}^r + u_{l,t}^r = d_{l,t}^r s_{l,t}^r, \quad \forall l \in [L], t \in [T], r \leq R, \quad (21e)$$

$$y_{k,l,t}^r, o_{k,t}^r, u_{l,t}^r \in \mathbb{R}_+, \quad \forall k \in [K], l \in [L], t \in [T], r \leq R. \quad (21f)$$

Record an optimal solution $(\mathbf{x}^*, \alpha^*, \beta^*, \delta^*)$ and set $LB = Z^*$.**Step 3: Solve sub-problem.**3.1 With $(\mathbf{x}, \alpha, \beta)$ fixed to be $(\mathbf{x}^*, \alpha^*, \beta^*)$, solve the following problem (22) for all $t \in [T]$.

$$W_t = \max_{\substack{\rho_l, \lambda_k, g_l, z_l, \\ v_l, r_l, q_l, e_l}} \left\{ \sum_{k=1}^K x_k h_k \lambda_{k,t} - \sum_{l=1}^L \left[\underline{d}_{l,t} \alpha_{l,t} + \Delta d_{l,t} g_{l,t} \alpha_{l,t} + \underline{s}_{l,t} \beta_{l,t} + \Delta s_{l,t} z_{l,t} \beta_{l,t} \right] \right. \\ \left. + \sum_{l=1}^L \left[\underline{d}_{l,t} \underline{s}_{l,t} \rho_{l,t} + \Delta s_{l,t} \underline{d}_{l,t} b_{l,t} + \Delta d_{l,t} \underline{s}_{l,t} q_{l,t} + \Delta d_{l,t} \Delta s_{l,t} e_{l,t} \right] : (20b)-(20l) \right\}. \quad (22)$$

3.2 Record the optimal solution (g_t^*, z_t^*) and optimal value W_t^* .3.3 Update $UB = \min\{UB, \sum_{t=1}^T W_t^* + (LB - \sum_{t=1}^T \delta_t^*)\}$.3.4 If $\frac{UB-LB}{UB} \leq \varepsilon$ stop and return $\mathbf{x}^*$ as the optimal solution of (18); else go to step 4.**Step 4. Column-and-Constraint Generation Routine**4.1 Use g_t^* to compute $d_{l,t}^{R+1} = \underline{d}_{l,t} + (\bar{d}_{l,t} - \underline{d}_{l,t})g_t^*$ and z_t^* to compute $s_{l,t}^{R+1} = \underline{s}_{l,t} + (\bar{s}_{l,t} - \underline{s}_{l,t})z_t^*$, for all $l \in [L]$ and $t \in [T]$.4.2 Add variables $y_{k,l,t}^{R+1}$, $u_{l,t}^{R+1}$, $o_{k,t}^{R+1}$, $\forall k \in [K], l \in [L], t \in [T]$, and the following constraints to the master problem:

$$\delta_t \geq \sum_{k=1}^K \sum_{l=1}^L c_{k,l,t} y_{k,l,t}^{R+1} + \sum_{k=1}^K c_{k,t}^o o_{k,t}^{R+1} + \sum_{l=1}^L c_{l,t}^u u_{l,t}^{R+1} - \sum_{l=1}^L (d_{l,t}^{R+1} \alpha_{l,t} + s_{l,t}^{R+1} \beta_{l,t}), \quad \forall t \in [T],$$

$$\sum_{l \in R_k} y_{k,l,t}^{R+1} + o_{k,t}^{R+1} = x_k h_k, \quad \forall k \in [K], t \in [T],$$

$$\sum_{k: l \in R_k} y_{k,l,t}^{R+1} + u_{l,t}^{R+1} = d_{l,t}^{R+1} s_{l,t}^{R+1}, \quad \forall l \in [L], t \in [T],$$

$$y_{k,l,t}^{R+1}, o_{k,t}^{R+1}, u_{l,t}^{R+1} \in \mathbb{R}_+, \quad \forall k \in [K], l \in [L], t \in [T].$$

update $R \leftarrow R + 1$, $\mathbf{O} \leftarrow \mathbf{O} \cup \{R + 1\}$ and go to step 2.**Table 2**HSCP Instances. L is # of service types, T is # of days, and K # of providers types.

Instance	L	K	T	Instance	L	K	T	Instance	L	K	T
1	4	4	30	6	4	6	180	11	6	6	90
2	4	4	90	7	4	8	30	12	6	6	180
3	4	4	180	8	4	8	90	13	6	8	30
4	4	6	30	9	4	8	180	14	6	8	90
5	4	6	90	10	6	6	30	15	6	8	180

we employed a widely used cross-validation method to estimate the optimal ϵ^* that minimizes the out-of-sample cost (see more details in Appendix E). We determine the sample size to use in the SAA formulations of the SP models that provides a good trade-off between solution time and approximation accuracy using a Monte Carlo Optimization (MCO) algorithm presented in Appendix F.1. Finally, we use

the same generated samples from LogN to obtain solutions to the SP and 1-Wasserstein DRO models.

According to staffing standards for various types of elderly care institutions in China, the minimum staffs number for serving less than 100 customers is three or five. Hence, we set the minimum number of providers, $\underline{w}$, to 3. We implement all models and solution methods

Table 3
The daily demand information from Pinetree Care.

Statistics	Rehabilitation nursing	Health status assessment
Mean, $\mu_{i,t}^d$	260	15
Support, $\mathcal{U}^d$	[124,398]	[11,19]
Standard deviation, $\sigma_{i,t}^d$	171.4	4.3

using the Python 3.8.8 programming language calling Gurobi V9.1.2 as a solver with default settings. We ran all experiments on a computer with Intel(R) Xeon(R) E3-1231-v3 processor with two 3.40 GHz CPUs and 32 GB shared RAM memory.

6.2. Obtaining near-optimal solutions to the SP models

For each instance in Table 2, we implemented the MCO algorithm in Appendix F.1 with $N_o = 10$, $N' = 1000$ and $K = 10$. Table F.1 in Appendix F.2 and Table F.10 in Appendix F.3 present the approximate optimality index ($AOI_N = \frac{\bar{v}_{N'} - \bar{v}_N}{\bar{v}_{N'}}$) and the 95% confidence intervals (95%CI) of the statistical lower and upper bounds ($\bar{v}_N$ and $\bar{v}_{N'}$, respectively) on the objective values of the E-SP and F-SP models for each instance under $(c_{i,t}^u, c_{i,t}^o, c_{k,t}^o) = (20, 2)$ and truncated LogN distribution. We observe similar results under different choices of these cost parameters. In the next section, we present and analyze the ranges of solution time of the SAA formulations. Herein we summarize the key findings of the MCO algorithm. Clearly, $N = 160$ and $N = 40$ scenarios of the demand and service duration are sufficient to obtain a near-optimal solution to the E-SP and F-SP models via their SAA formulations, respectively. First, AOI_N at termination sample size ranges from 0 to 0.0092 and from 0 to 0.0061 for the E-SP and F-SP models, respectively. Second, the 95%CI of $\bar{v}_N^E$ ($\bar{v}_N^F$) and $\bar{v}_{N'}^E$ ($\bar{v}_{N'}^F$) with $N = 160$ (40) are very tight (i.e., have a small variance). These results qualify $\bar{v}_N^E$ and $\bar{v}_N^F$ as tight statistical estimates for the lower and upper bounds on the optimal value of each instance, respectively. In addition, we also use truncated normal distribution and uniform distribution to generate the sample data. We obtain similar conclusion for these distributions (see Appendices F.2 and F.3).

It is worth mentioning that solving these models with larger N resulted in longer solution times without consistent and significant improvements in the relative approximation gaps. However, as shown in Appendices F.2 – F.3, solution times are still reasonable under $N = 500$ scenarios, and the AOI_N values similar to those obtained under the selected sample sizes.

6.3. CPU time and computational details

In this section, we analyze the solution times of the proposed SP and DRO models. In addition to the base range of the demand described in Section 6.1, we consider another range [40, 100], which corresponds to a higher amount of demand and variation level. For each of the 15 instances in Table 2 and demand range, we generated 10 instances as described in Section 6.1 for a total of 300 random instances. Then, we solved these instances using the proposed models for each decision-maker.

Let us first compare the computational performance of the SP models. Table 4 presents the computational details (i.e., the minimum (Min), average (Avg), and maximum (Max) CPU time in seconds) for E-SP and F-SP models. We observe the following from this table. First, the computational effort tends to increase with the scale ($L \times K \times T$). Second, both models can quickly solve all instances to optimality within 60 s. Third, solution times of F-SP models are much longer than the E-SP. For example, the average solution time of instance 15 (largest instance) using the E-SP and F-SP models under demand range [40, 60] are 8.6 and 22.7 s, respectively. These results make sense because the F-SP model has a larger second stage than the E-SP model.

Let us now analyze solution times of DRO models. Table 5 presents the computational details for E-MDHSCP and F-MDHSCP models. Table 6 presents the computational details for E-WDHSCP and F-WDHSCP models with $N = 10$. For brevity, in Appendix G.1, we present the computational details for E-WDHSCP and F-WDHSCP models with $N = 50$ and $N = 100$ in Table G.1. We observe the following from these tables. First, DRO models based on mean-support ambiguity set take a shorter time than the SP and the Wasserstein-based models for both EA and FA under the two ranges of the demand. This makes sense because the mean-support DRO model is a deterministic model and does not scale with the sample size.

Second, the E-WDHSCP model with $N = 10$ has similar solution times as the E-SP model. However, the E-WDHSCP models with $N = 50$ and $N = 100$ take much longer times than E-SP model. Note that solution times of the DRO models based on 1-Wasserstein ambiguity set increase with the sample size. This makes sense because the number of decision variables and constraints of these DRO models increases with N . Third, the F-WDHSCP models take much longer to solve each instance than the F-SP model. The average solution time of the F-WDHSCP model with $N = 10$ and $N = 50$ ranges from 6.6 to 103.4 and from 32.1 to 862.3 s, respectively.

Finally, we also solve the E-WDHSCP and F-WDHSCP models with the terminated sample size obtained using MCO for the SAA models. The results in Appendix G.2 show that the E-WDHSCP model can solve these instances within 610 s, and the F-WDHSCP model can solve all of them within 600 s. Overall, these results demonstrate the computational efficiency of the proposed approaches. To confirm our conclusion for the scenario-based SAA models, we solve the SAA formulations of E-SP and F-SP models with 500 samples. We solve 6 instances with $L = 6$ under these settings and present solution times in Tables F.3 (see Appendix F.2) and Table F.12 (see Appendix F.3). Clearly, the E-SP model can solve these instances within 30 s, and the F-SP model can solve all of them within 30 min.

6.4. Efficiency of valid inequalities

To investigate the efficiency of the proposed valid inequalities (24)–(25) for the F-MDHSCP, we separately solve this model via the proposed C&CG with (w/) and without (w/o) inequalities (24)–(25).

Table 7 presents the computational details (CPU time, number of iterations in the C&CG algorithm before it converges to the optimum, and gap) of solving F-MDHSCP w/ and w/o inequalities (24)–(25). It is clear that the computational effort increases w/o inequalities (24)–(25). Specifically, the algorithm takes hundreds of iterations and longer solution time to solve each instance w/o inequalities (24)–(25). Moreover, for most instances, the algorithm w/o these inequalities terminates at the specified time limit (i.e., 7200 s) with a large optimality gap ($\frac{\text{Upper Bound} - \text{Lower Bound}}{\text{Upper Bound}}$) of around 0.99. In contrast, the algorithm takes a few seconds (i.e., around 10 s) and iterations (i.e., around 6 iterations) to solve each instance w/ inequalities (24)–(25).

Given that we observe similar results for all instances, for brevity and illustrative purposes, we use Instance 15 ($L = 6, K = 8, T = 180$) under demand range 1 to present the convergence details. Recall that the algorithm takes hundreds of iterations and a longer time until convergence w/o inequalities (24)–(25). Therefore, in Figs. 2(a)–2(b), we present the lower bound and gap values w/ and w/o inequalities (24)–(25) from the first 20 iterations. It is obvious that both the lower bound and gap values converge faster when we introduce inequalities (24)–(25) into the master problem. Moreover, because of the better bounding effect, the algorithm converges to the optimal solution in fewer iterations and shorter solution times.

Table 4
CPU times in seconds for SP models.

Inst	E-SP						F-SP					
	Demand: [40, 60]			Demand: [40, 100]			Demand: [40, 60]			Demand: [40, 100]		
	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
1	0.8	0.8	0.9	0.8	0.9	1.0	0.8	0.9	0.9	0.8	0.9	1.1
2	2.5	2.7	3.0	2.5	2.6	3.1	3.0	3.4	3.9	3.1	3.4	3.9
3	5.0	5.1	5.2	5.3	5.5	6.3	9.6	10.0	10.4	10.3	11.4	13.1
4	0.9	0.9	1.0	0.9	0.9	1.0	1.0	1.0	1.2	1.3	1.5	1.7
5	2.6	2.8	3.0	2.7	2.8	3.0	3.6	4.0	5.9	7.9	8.3	9.4
6	5.5	6.1	7.2	5.3	5.6	6.0	10.4	11.2	12.3	15.3	17.3	21.3
7	0.9	0.9	1.0	0.8	0.9	0.9	1.6	1.8	2.0	1.6	1.8	2.2
8	2.7	2.9	3.0	2.7	2.8	2.9	8.5	9.8	13.1	7.5	9.0	10.9
9	5.8	5.9	6.1	5.5	5.6	5.7	27.2	34.7	49.7	24.3	26.7	29.6
10	1.2	1.3	1.3	1.2	1.3	1.3	1.7	1.9	2.1	1.6	1.7	1.9
11	3.8	3.9	4.0	3.9	4.0	4.2	7.9	9.2	13.2	7.7	8.8	12.0
12	8.0	8.1	8.3	8.1	8.2	8.4	24.1	30.1	52.6	23.3	28.1	38.1
13	1.3	1.4	1.7	1.3	1.3	1.4	1.9	2.1	2.3	1.9	2.1	2.3
14	4.0	4.3	4.5	4.0	4.1	4.3	7.9	9.0	12.9	8.1	9.2	12.8
15	8.2	8.6	9.2	8.3	8.8	9.4	21.0	22.7	29.4	21.7	25.0	33.1

Table 5
CPU times in seconds for mean-support based DRO models.

Inst	E-MDHSCP						F-MDHSCP					
	Demand: [40, 60]			Demand: [40, 100]			Demand: [40, 60]			Demand: [40, 100]		
	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
1	0.1	0.1	0.1	0.1	0.1	0.1	0.5	0.5	0.6	0.6	0.6	0.7
2	0.2	0.2	0.3	0.2	0.2	0.3	1.4	1.6	1.9	1.3	1.5	1.7
3	0.5	0.5	0.5	0.4	0.5	0.5	2.8	2.8	2.9	4.0	4.1	4.2
4	0.2	0.2	0.2	0.1	0.1	0.1	0.9	0.9	1.0	1.0	1.1	1.2
5	4.4	4.6	5.0	0.2	0.3	0.3	2.2	2.2	2.2	5.0	5.2	5.3
6	5.5	5.7	5.9	0.5	0.5	0.6	3.0	3.1	3.3	9.3	10.9	13.9
7	0.1	0.1	0.1	0.1	0.1	0.1	1.0	1.0	1.1	1.2	1.2	1.3
8	0.3	0.3	0.4	0.2	0.2	0.3	3.4	3.5	3.6	3.6	3.7	3.7
9	0.6	0.6	0.7	0.5	0.5	0.6	3.8	3.9	4.0	11.2	12.3	16.2
10	0.1	0.1	0.1	0.1	0.1	0.1	1.8	1.8	1.9	1.1	1.1	1.2
11	0.3	0.3	0.4	0.3	0.3	0.4	2.7	2.8	2.8	1.9	2.0	2.1
12	0.7	0.7	0.7	0.7	0.7	0.8	4.1	4.2	4.3	4.0	4.1	4.1
13	0.1	0.1	0.2	0.1	0.1	0.1	1.0	1.0	1.1	0.7	0.7	0.8
14	0.4	0.5	0.5	0.4	0.4	0.4	4.8	4.9	5.1	8.5	8.6	8.7
15	1.5	1.6	1.7	0.8	0.8	0.9	4.7	4.8	5.0	10.4	10.5	10.6

Table 6
CPU times in seconds for 1-Wasserstein distance based DRO models (sample size $N = 10$).

Inst	E-WDHSCP						F-WDHSCP					
	Demand: [40, 60]			Demand: [40, 100]			Demand: [40, 60]			Demand: [40, 100]		
	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max	Min	Avg	Max
1	0.8	0.8	0.9	0.7	0.8	0.8	6.6	7.0	7.8	9.8	10.9	11.9
2	2.4	2.5	2.6	2.4	2.6	2.7	22.3	25.3	30.6	29.5	34.0	38.4
3	5.0	5.1	5.2	4.8	4.9	4.9	52.6	54.6	56.7	43.8	45.6	48.5
4	1.1	1.2	1.3	1.1	1.1	1.2	7.3	7.8	11.5	10.5	10.9	11.4
5	3.1	3.2	3.6	3.1	3.2	3.3	24.2	24.7	25.6	22.6	27.8	38.6
6	5.2	5.5	5.6	5.5	5.5	5.6	56.5	63.5	69.2	51.4	52.5	54.4
7	1.3	1.3	1.4	1.2	1.3	1.3	7.7	8.1	8.8	11.5	12.1	13.0
8	3.6	3.7	3.8	3.7	3.7	3.9	29.8	30.5	31.6	27.6	38.1	49.9
9	6.2	6.5	8.8	6.9	7.0	7.8	90.9	97.2	103.4	67.0	71.6	76.4
10	1.2	1.3	1.3	1.1	1.2	1.2	9.6	9.9	10.5	13.6	14.6	16.0
11	3.5	3.6	3.7	3.4	3.4	3.5	32.7	35.1	36.8	32.5	33.8	36.0
12	7.6	7.8	8.4	7.1	7.4	7.6	84.5	88.5	94.0	71.4	79.2	83.1
13	1.6	1.7	1.8	1.6	1.6	1.7	10.8	11.0	11.5	15.3	16.0	16.8
14	4.0	4.1	4.4	4.1	4.1	4.3	36.1	37.1	38.7	33.9	35.1	36.9
15	8.2	8.3	8.4	8.0	8.1	8.2	88.7	97.0	102.7	80.9	84.2	88.1

6.5. Analysis of results of EA models

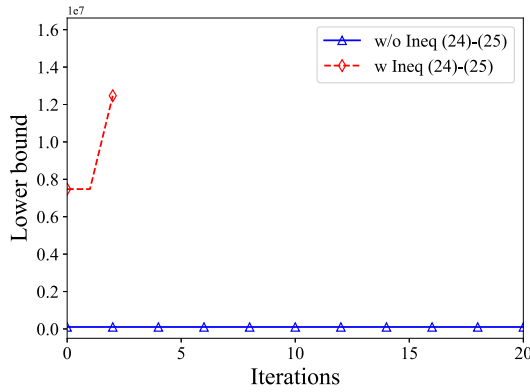
In this section, we analyze the optimal staffing patterns of the E-SP, E-MDHSCP, and E-WDHSCP models (Section 6.5.1) and their out-of-sample performance (Section 6.5.2). To analyze solutions to the problem under limited data and distributional information, we obtain solutions to the E-SP and E-WDHSCP models using a sample of $N = 10$ scenarios of the demand and service times generated as described

in Section 6.1. Using a larger sample size did not lead to different decisions. For illustrative purposes and brevity, we present results for the real case instance ($L = 2, K = 3, T = 30$), which is constructed based on the data from Pinetree. In addition, we consider three types of providers in real case, i.e., $K = 3$. We let providers types $k \in \{1, 2\}$ represent *specialized* providers (i.e., they can only provide one type of service), where $k = 1$ and $k = 2$ are specialized providers in rehabilitation nursing and health status assessment, respectively. And

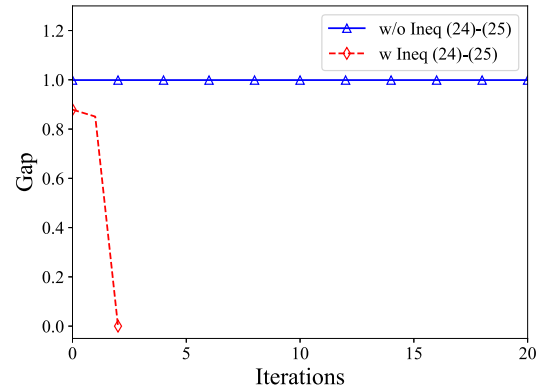
Table 7

Iterations (Iters), CPU time, and gap values of F-MDHSCP w/ and w/o inequalities (24)–(25).

Inst	Demand range: [40, 60]						Demand range: [40, 100]					
	w/o Ineq (24)–(25)			w/ Ineq (24)–(25)			w/o Ineq (24)–(25)			w/ Ineq (24)–(25)		
	Iters	Time	Gap	Iters	Time	Gap	Iters	Time	Gap	Iters	Time	Gap
1	162	484	0.0	3	0.5	0.0	191	671	0.0	3	0.6	0.0
2	292	7200	1.0	3	1.6	0.0	342	7200	1.0	4	1.4	0.0
3	174	7200	1.0	3	3.3	0.0	246	7200	1.0	4	4.2	0.0
4	178	618	0.0	3	0.6	0.0	214	972	0.0	4	1.1	0.0
5	331	7200	1.0	3	2.2	0.0	341	7200	1.0	4	5.0	0.0
6	161	7200	1.0	3	3.3	0.0	228	7200	1.0	4	9.5	0.0
7	67	172	0.0	3	0.8	0.0	211	1145	0.0	4	1.4	0.0
8	230	7200	1.0	3	3.0	0.0	184	7200	1.0	3	2.9	0.0
9	126	7200	1.0	3	3.9	0.0	233	7200	1.0	4	8.1	0.0
10	196	3776	0.0	3	0.9	0.0	240	7200	1.0	3	1.0	0.0
11	121	7200	1.0	3	2.4	0.0	134	7200	1.0	3	3.2	0.0
12	79	7200	1.0	3	4.9	0.0	77	7200	1.0	4	4.8	0.0
13	324	3583	0.0	3	1.2	0.0	342	4105	0.0	3	1.0	0.0
14	271	7200	1.0	3	4.6	0.0	286	7200	1.0	6	8.4	0.0
15	191	7200	1.0	3	4.9	0.0	200	7200	1.0	4	9.6	0.0



(a) LB values



(b) Gap values

Fig. 2. Comparisons of lower bound and gap values of F-MDHSCP with and without valid inequality.

we let type $k = 3$ represents *cross-trained* providers who are trained or qualified for two types of services.

6.5.1. Optimal staffing patterns of EA models

In this section, we compare the optimal staffing pattern (i.e., the optimal number of hired providers) of the E-SP, E-MDHSCP, and E-WDHSCP models under different values of the under-staffing $c_{l,t}^u$ and over-staffing $c_{l,t}^o$ penalty costs. First, in Fig. 3(a), we present the optimal number of hired providers for the real case with $c_{l,t}^o = 1$ and $c_{l,t}^u \in \{1, 5, 10, 15, 20, 25\}$. We observe the following from this figure. First, it is clear that all models hire more providers as the under-staffing cost increases from 1 to 25. However, the number of hired providers start to stabilize after $c_{l,t}^u = 10$. This makes sense because, as we later show, when $c_{l,t}^u = 10$ and $c_{l,t}^o = 1$, under-staffing is too low (see Section 6.5.2 for more details). Hence, increasing the number of providers will increase the fixed employment cost without significantly improving operational performance and the associated cost. Second, when $c_{l,t}^u < 5$ (i.e., low under-staffing cost), the E-SP and E-WDHSCP models hire the same number of providers, and when $c_{l,t}^u = 5$ the E-MDHSCP model hire fewer providers. In contrast, when $c_{l,t}^u > 10$, the E-MDHSCP models hire more providers than the other models to mitigate excessive under-staffing. Third, E-WDHSCP models hire more providers than the E-SP models when $c_{l,t}^u > 10$. These results indicate that $c_{l,t}^u = 10$ may be a suitable under-staffing penalty cost for the EA decision-makers. We observe a different pattern when we fix $c_{l,t}^u = 10$ and increase the penalty for over-staffing $c_{l,t}^o$ from 1 to $\{5, 10, 15, 20, 25\}$. Specifically, as shown in Fig. 3(b) both models hire less providers as $c_{l,t}^o$ increases. And the number of hired providers stabilize after $c_{l,t}^o = 10$. In

fact, when $c_{l,t}^o \geq 10$, the number of hired providers is close to the lower bound on the number of required providers.

Next, we analyze the detailed staffing patterns under cost structure 1: $(c_{l,t}^u, c_{l,t}^o) = (10, 1)$, and cost structure 2: $(c_{l,t}^u, c_{l,t}^o) = (10, 10)$ presented in Fig. 4. First, under the two cost structures, the E-MDHSCP model always prefers to hire specialized providers to serve customers (e.g., the E-MDHSCP model did not hire any of the cross-trained providers). Second, we observe that E-MDHSCP model hires more providers under cost structure 1, and fewer providers under cost structure 2 than E-SP and E-WDHSCP model. This result indicates that E-MDHSCP model focuses more on minimizing over-staffing as $c_{l,t}^o$ increases from 1 to 10. Third, E-WDHSCP model hires a slightly higher number of providers than E-SP model.

The results in this section show that different methods of uncertainty lead to different staffing patterns. In the next section, we analyze the impact of these solutions on operational performance.

6.5.2. Out-of-sample performance of EA's solutions

In this section, we compare the operational performance of the optimal solutions of the E-MDHSCP, E-WDHSCP, and E-SP via out-of-sample simulation. We test the out-of-sample performance (i.e., the objective value obtained by simulating the optimal solution of a model under a larger unseen data) as follows. First, we solve each instance and obtain the optimal first-stage decisions. Second, we fix the obtained first-stage decisions in the second stage of the SP model, then re-optimize the second stage of the SP using the following sets of $N' = 10,000$ out-of-sample data to compute the second stage cost (i.e., capacity allocation cost, under-staffing plus over-staffing costs).

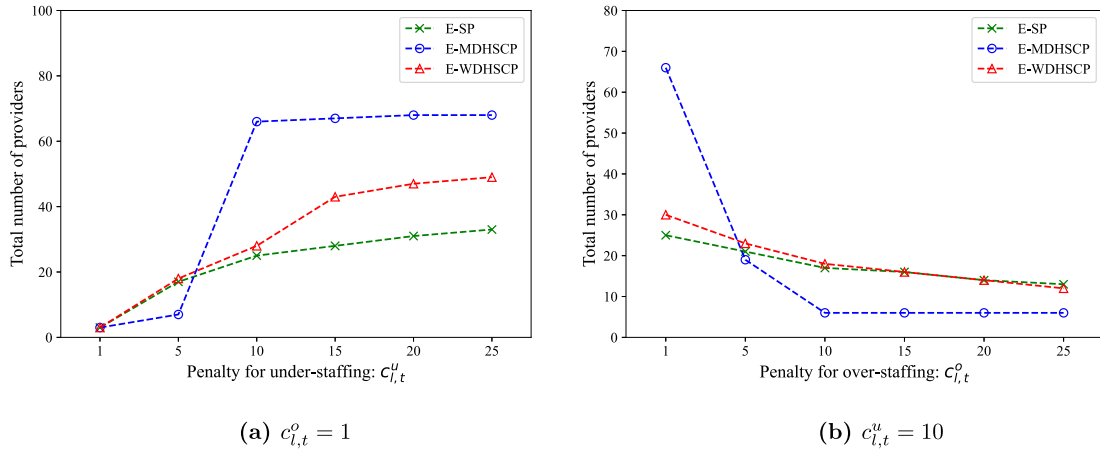


Fig. 3. Total number of providers hired by SP and DRO models for EA.

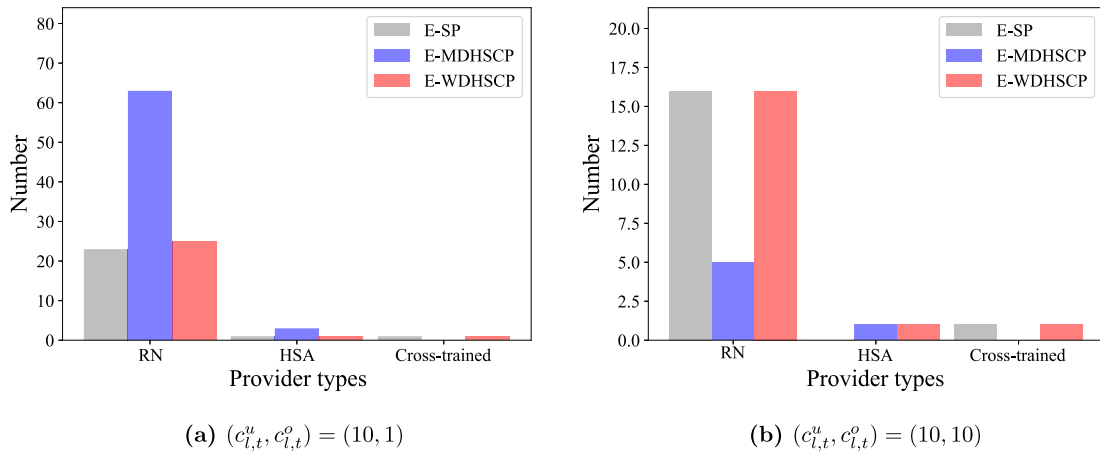


Fig. 4. Staffing patterns of EA decision-makers for the real case.

We compute the out-of-sample total cost as first-stage cost plus the out-of-sample second-stage cost.

- Set 1. We assume that we have perfect distributional information. That is, we use the same distributions (LogN) and parameter settings that we use for generating the N data in the optimization to generate N' data. This is to simulate the performance when the true distribution is the same as the one used in the optimization.
- Set 2. In this set, we assume that we have used misspecified distributions of the demand and service duration in the optimization (motivated by our analysis in Section 2). We follow a similar out-of-sample simulation testing procedure described in Shehadeh (2023) and Wang et al. (2020) and perturb the support of the random demand and service durations by a parameter Δ and used parameterized uniform distribution $U[(1 - \Delta) \text{ lower bound}, (1 + \Delta) \text{ upper bound}]$ to generate the $N' = 10,000$ data. We apply $\Delta \in \{0, 0.1, 0.25, 0.5\}$, where a higher Δ indicate a larger variation.

For brevity, we again focus on real case, and present results under cost structure 1, i.e., $(c_{l,t}^u, c_{l,t}^o) = (10, 1)$, (we obtain similar observations for the other instances). First, let us analyze simulation results under Set 1 (i.e., perfect distributional information case). Fig. 5 presents histograms of the out-of-sample total, second-stage, and under-staffing costs. It is clear from these figures that DRO models yield lower under-staffing costs than E-SP model on average and at all quantiles. This makes sense because the DRO models tends to hire more providers to

hedge against under-staffing. However, this may result in a higher fixed cost associated with hiring more providers. E-WDHSCP model yields a lower second stage cost than other two models on average and at all quantiles.

We observe the following from simulation results under cost structure 1 and Set 2 (i.e., misspecified distributional information case) presented in Fig. 6. First, the DRO models consistently outperform the E-SP model under all levels of variation (Δ) and across the criteria of mean and all quantiles of the second-stage cost and under-staffing cost. In particular, E-MDHSCP model yields a significantly lower under-staffing cost than other two models. Note that lower second-stage and under-staffing costs indicates better operational performance (i.e., lower shortage and meeting greater demand) and thus has a significant practical impact, especially in healthcare. Second, E-WDHSCP model always outperforms E-SP model in terms of the total cost, second stage cost, and under-staffing cost. Third, E-WDHSCP and E-SP models yield lower total cost than E-MDHSCP under low level variation. However, under a higher level of variation ($\Delta \geq 0.25$), E-MDHSCP yields a lower total cost. The superior performance of the DRO model reflects the value of modeling uncertainty and distributional ambiguity.

We also presents the results under cost structure 2, i.e., $(c_{l,t}^u, c_{l,t}^o) = (10, 10)$, in Figure H.1 (see Appendix H). We observe followings from this figure. First, E-WDHSCP and E-SP yield lower total, second stage, and under-staffing costs than E-MDHSCP under all levels of variation (Δ) and across the criteria of mean and all quantiles. This makes sense because under this cost structure, the E-MDHSCP hires fewer providers to mitigate the risk of high over-staffing cost, leading to significant under-staffing and hence higher second-stage costs. Second, E-WDHSCP

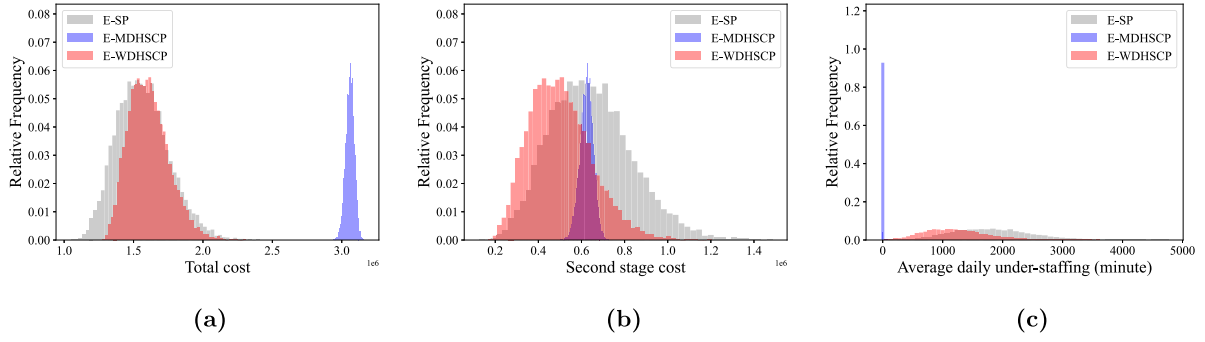


Fig. 5. Out-of-sample performance of EA models for real case under $(c_{I,t}^u, c_{I,t}^o) = (10, 1)$, Set 1.

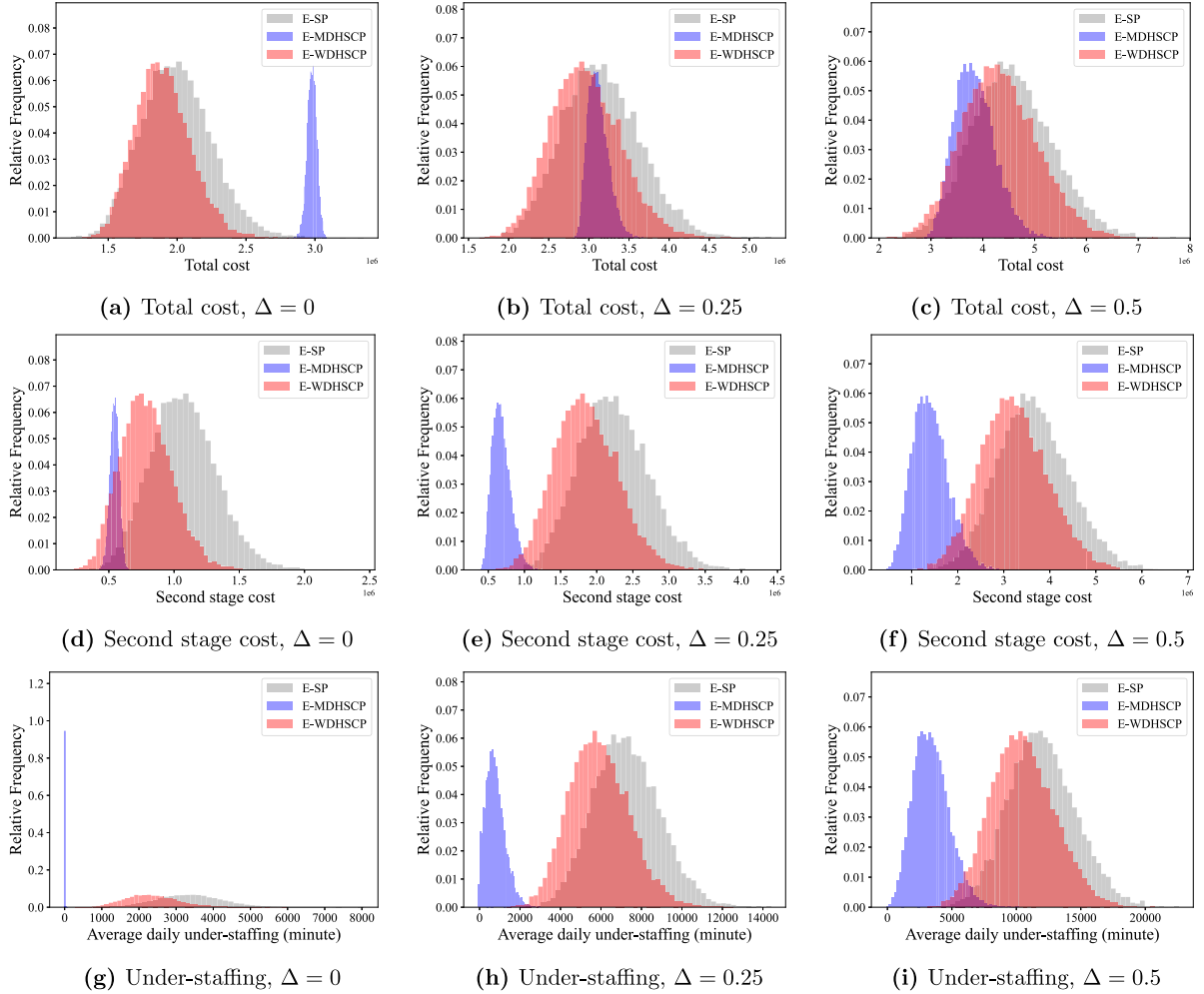


Fig. 6. Out-of-sample performance of EA models for real case under $(c_{I,t}^u, c_{I,t}^o) = (10, 1)$ and Set 2.

model yields slightly lower second stage cost and under-staffing cost than E-SP, indicating better operational performance. This also makes sense because, as we show in Section 6.5.1, the E-WDHSCP model hires a slightly higher number of providers than the E-SP model. Hence it mitigates under-staffing and leads to lower second-stage costs.

Next, we investigate the value of distributional robustness from the perspective of out-of-sample disappointment, which measures the extent to which the out-of-sample cost exceeds the model's optimal value (Van Parys et al., 2021; Wang et al., 2020). Following Shehadeh (2023) and Wang et al. (2020), we define the out-of-sample disappointment as $\max\left\{0, \frac{V_{out} - V_{opt}}{V_{opt}}\right\} \times 100\%$, where V_{opt} and V_{out} are as the model's optimal value and the out-of-sample objective value, respectively. A

disappointment of zero indicates that the model's optimal value is equal to or larger than the out-of-sample (actual) cost, suggesting that the model is more robust and avoids underestimating costs. In contrast, a larger disappointment implies a higher level of over-optimism because, in this case, the actual cost (V_{out}) of implementing the optimal solution of a model is larger than the estimated cost (V_{opt}).

Fig. 7 presents histograms of the out-of-sample disappointment. The estimated cost, V_{opt} , for E-SP, E-MDHSCP, and E-WDHSCP models are 1564547, 3161325, and 2156595, respectively. While the estimated costs of DRO models are significantly higher than that of SP, the results in Fig. 7 indicate that the estimates of the actual cost provided by the DRO model are more accurate since they lead to lower disappointment

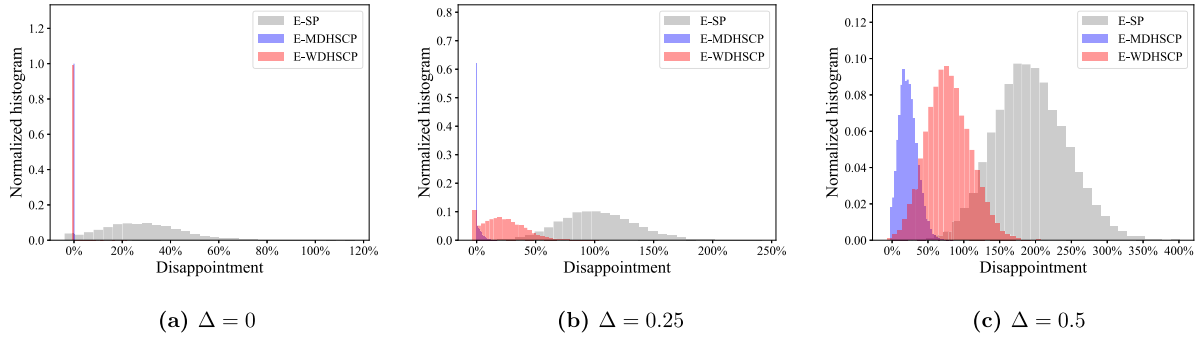


Fig. 7. Out-of-sample disappointment of EA models for real case under $(c_{l,t}^u, c_{k,t}^o) = (10, 1)$, Set 2.

than the SP model. Specifically, the DRO models yield smaller out-of-sample disappointments on average and at all quantiles than the E-SP solutions under all levels of variation (Δ). Among the two DRO models, the E-MDHSCP model yields smaller out-of-sample disappointments on average and at all quantiles than the E-WDHSCP model. Moreover, the range of E-MDHSCP disappointments is stable. In contrast, the disappointment of the E-SP model significantly increases under higher variation. For example, the disappointment of the E-MDHSCP solution is smaller than 50% under a higher variation $\Delta = 0.5$ while the disappointment of the E-SP model significantly increases, exceeding 200%. These results confirm that the DRO models provide more robust estimates of the actual cost that we will be observed in practice. In a decision-making context where the goal is to minimize expenses, disappointments are more harmful than positive surprises (Van Parys et al., 2021).

6.6. Analysis of results of FA models

In this section, we analyze the optimal staffing patterns of the F-SP, F-MDHSCP, and F-WDHSCP models (Section 6.6.1) and their out-of-sample performances (Section 6.6.2). We obtain solutions to the SP and F-WDHSCP models using a sample of $N = 10$ scenarios of demand and service time generated as described in Section 6.1. We again present results for the real case defined in Section 6.5.

6.6.1. Optimal staffing patterns of FA models

In this section, we compare the optimal staffing pattern of the F-SP, F-MDHSCP, and F-WDHSCP models under different values of the under-staffing $c_{l,t}^u$ and over-staffing $c_{k,t}^o$ penalty costs. In Fig. 8, we present the optimal number of hired providers for the real case with $c_{k,t}^o = 1$ and $c_{l,t}^u \in \{1, 2, 3, 4, 5, 6, 7\}$. We observe the following from these figures. First, it is clear that all models hire more providers as the under-staffing cost increases from 1 to 7. However, the number of hired providers start to stabilize after $c_{l,t}^u \geq 5$. When $c_{l,t}^u \geq 5$, the F-MDHSCP models hire more providers than the F-SP and F-WDHSCP models to mitigate the risk of excessive under-staffing. Second, when $c_{l,t}^u \leq 3$ (i.e., low under-staffing cost), all models hire approximately the same number of providers. These results indicate that $c_{l,t}^u = 5$ may be a suitable penalty cost of under-staffing for FA decision-makers. We observe a different pattern when we fix $c_{l,t}^u = 5$ and increase the penalty for over-staffing $c_{k,t}^o$ from 1 to $\{2, 3, 4, 5, 6, 7\}$. Specifically, as shown in Fig. 8(b), all models hire less providers as $c_{k,t}^o$ increases, which makes sense because the cost of over-staffing is higher in this setting. The number of hired providers stabilize after $c_{k,t}^o = 2$.

Next, we analyze the detailed staffing patterns for the real case presented in Fig. 9. First, F-MDHSCP model hires more providers under cost structure $(c_{l,t}^u, c_{k,t}^o) = (5, 1)$ than under cost structure $(c_{l,t}^u, c_{k,t}^o) = (5, 5)$. Again, this indicates that F-MDHSCP model focus more on avoiding over-staffing when $c_{k,t}^o$ increases from 1 to 5. Second, F-WDHSCP model hires all types of providers under the two cost structures.

6.6.2. Out-of-sample performance of FA model

In this section, we compare the operational performance of the optimal solutions of the F-MDHSCP, F-WDHSCP, and F-SP via out-of-sample simulation testing. Using the same methods in Section 6.5.2, we generate two sets of $N' = 10,000$ out-of-sample data (Set 1-2). For brevity, we again focus on real case (we obtain similar observations for the other instances). Let us analyze simulation results under $(c_{l,t}^u, c_{k,t}^o) = (5, 1)$ and Set 1. Figure I.1 in Appendix I.1 presents histograms of the out-of-sample total, second-stage, and under-staffing costs. We observe followings from this figure. First, F-MDHSCP model yields a lower under-staffing cost than F-SP and F-WDHSCP models. This makes sense because the F-MDHSCP hires more providers than the other two models; hence it mitigates under-staffing. Second, F-WDHSCP yields slightly lower second stage cost and under-staffing cost than F-SP model on average and at all quantiles.

Next, we observe the following from the simulation results under $(c_{l,t}^u, c_{k,t}^o) = (5, 1)$ and Set 2 for the real case presented in Figure I.2 (see Appendix I.1). First, F-MDHSCP model yields significantly lower under-staffing than other two models under all high levels of variation and across the criteria of mean and all quantiles. This is because F-MDHSCP model hires a larger number of providers. However, this also leads to a higher total cost. Second, F-WDHSCP model yields slightly lower total cost, second stage cost, and lower under-staffing cost than F-SP model, i.e., indicating better operational performance. Third, for high levels of variation, i.e., $\Delta \geq 0.25$, F-MDHSCP model yields significantly lower second stage cost than other two models. Again, the superior performance of the DRO model reflects the value of modeling uncertainty and distributional ambiguity.

Then, in Figure I.3 (see Appendix I.2), we present the results under $(c_{l,t}^u, c_{k,t}^o) = (5, 5)$ for the real case. We observe the following from this figure. First, F-SP and F-WDHSCP models yield lower total cost, second-stage costs and under-staffing than the F-MDHSCP model under across the criteria of mean and all quantiles. Recall that F-MDHSCP hires a significantly lower number of providers under cost structure this cost structure than other two models. Therefore, it leads to higher under-staffing and second stage cost. Second, F-WDHSCP model yields a slightly lower under-staffing than the F-SP model, i.e., indicating better operational performance.

Fig. 10 presents histograms of the out-of-sample disappointment under this cost structure. It is clear from this figure that the DRO models yield substantially smaller out-of-sample disappointments on average and at all quantiles than the F-SP model. These results confirm again that the DRO models provide more robust estimates of the actual cost.

6.7. Comparisons of EA and FA solutions

So far, we have focused our analysis on optimal solutions to the proposed models for each type of decision-maker. While these models are not equivalent, in this section, we attempt to compare the performance of solutions to the FA and EA models. Recall that we have shown the

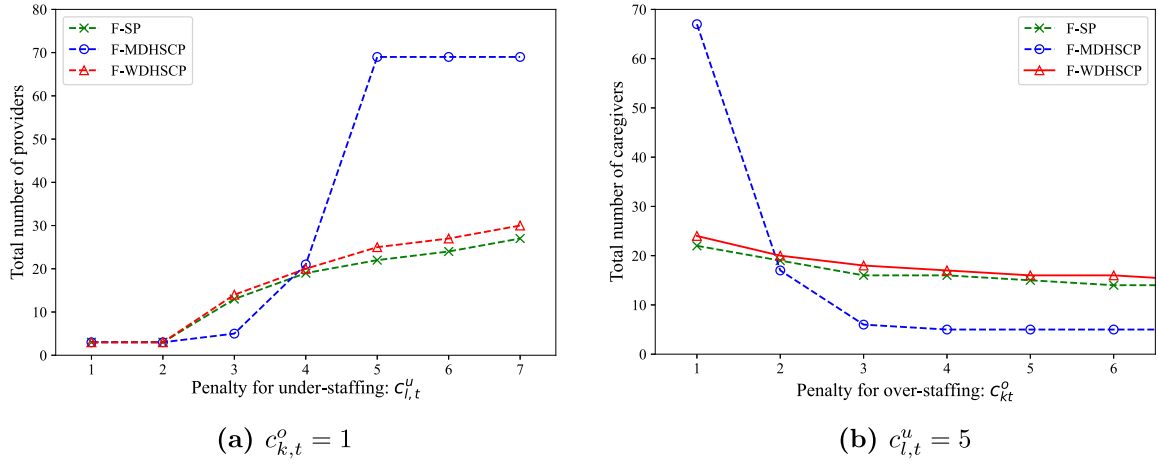


Fig. 8. Total number of providers hired by F-SP, F-MDHSCP, and F-WDHSCP models for real case.

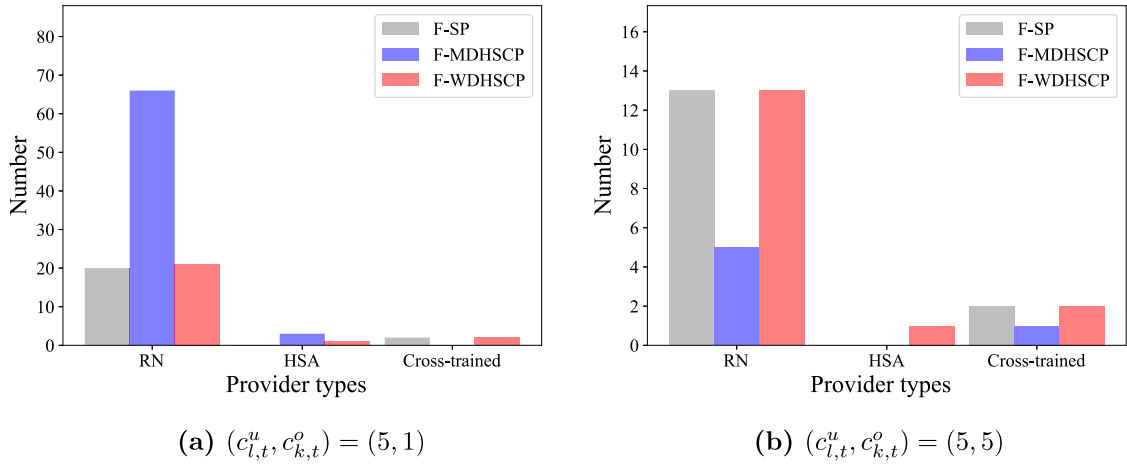


Fig. 9. Staffing patterns of FA decision-makers for real case.

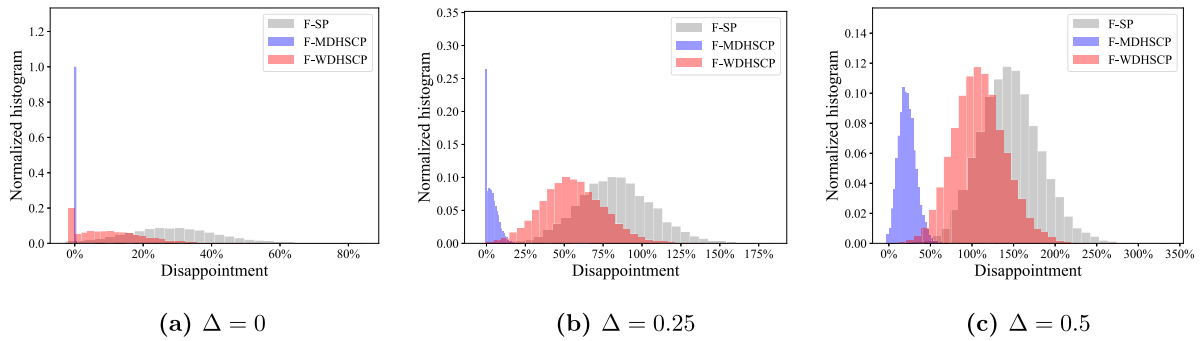


Fig. 10. Out-of-sample disappointment of FA models for real case under Set 2.

superior performance of the DRO models in Sections 6.5–6.6. Therefore, we focus on the comparisons of DRO models in this section. For brevity, we again focus on real case instance. Furthermore, we consider an equal cost structure, $(c_{l,t}^u, c_{l,t}(c_{k,t}^o)) = (10, 10)$, which provide a fair comparison to a certain extent. Finally, for E-WDHSCP and F-WDHSCP models, we use the same settings in Sections 6.5 and 6.6, respectively.

Fig. 11 presents the optimal staffing patterns of four DRO models. We observe that EA and FA models that employ the same type of ambiguity set hire approximately the same number of providers. This could indicate that under the same ambiguity set, the decision-maker type does not significantly impact the total number of hired providers. On the other hand, the decision-maker types impact staffing patterns. For example, the FA models hire more cross-trained providers (type 3)

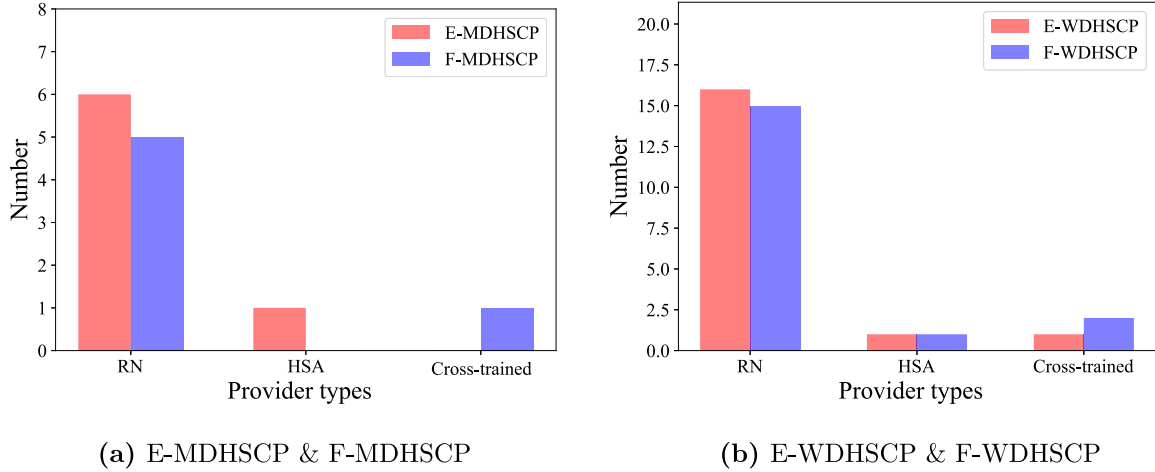


Fig. 11. Staffing patterns of EA models and FA models.

than the EA models, and the EA models hire more specialized providers (types 1 and 2). Recall that FA models determine capacity allocations based on demand realizations. Thus, it could provide more flexibility in the allocation decisions by hiring more cross-trained providers who can provide both RN and HSA services.

Fig. 12 compares the out-of-sample total cost and under-staffing under Set 2 with $\Delta = 0$. It is clear from this figure that E-MDHSCP solution yields slightly lower total cost and under-staffing cost than F-MDHSCP solution. In contrast, the E-WDHSCP and F-WDHSCP models yield approximately the same total and under-staffing costs. Thus, there is no clear winner between these models in terms of cost. This makes sense because EA and FA models hire approximately the same number of providers.

Finally, we remark that the DRO models for the EA decision-maker can be implemented and efficiently solved using off-the-shelf optimization software, enabling practitioners to use these models directly. In contrast, the DRO models for the FA decision-maker require specially-developed algorithms (C&CG). Thus, we claim that the EA models have the advantage of implementability, i.e., they can be easily translated into standard optimization software packages, not requiring customized algorithmic development. As pointed out by Shehadeh et al. (2019), implementability in the above sense is *necessary for an optimization-based decision support tool to gain wide adoption in service-providing systems such as home care that do not have ongoing access to support staff with optimization expertise*, and thus is a valuable feature of the proposed DRO models for EA decision-maker models.

7. Conclusion

In this paper, we proposed stochastic optimization methodologies for the HSCP problem, which seeks to find the number of providers to hire and their allocation decisions to different types of services in each day within a specified planning horizon. The objective is to minimize the total cost associated with staffing (i.e., employment), capacity allocation, over-staffing, and under-staffing. To address uncertainty in service duration and the demand for home service, we proposed and analyzed SP and DRO models considering two types of decision-makers, namely everything in advance decision-maker (for which we proposed E-SP, E-MDHSCP, and E-WDHSCP models) and flexible adjustment decision-maker (for which we proposed F-SP, F-MDHSCP, and F-WDHSCP models). We derive equivalent MILP reformulations of the proposed nonlinear E-MDHSCP and E-WDHSCP models for the EA decision-maker that can be implemented and efficiently solved

using off-the-shelf optimization software. We propose a computationally efficient column-and-constraint generation algorithm with valid inequalities to solve the proposed F-MDHSCP and F-WDHSCP models for the FA decision-maker.

Finally, we conduct extensive numerical experiments comparing the operational and computational performance of the proposed approaches and discuss insights and implications for home service staffing and capacity planning. The results demonstrate (1) the DRO models tend to hedge against uncertainty and ambiguity by hiring more providers and thus have a higher fixed cost but significantly lower operational (i.e., under-staffing) costs than the SP models, (2) the DRO solutions are more robust under various distributions with substantially lower disappointments and thus higher reliability than the SP models, (3) the computational efficiency of the proposed approaches, indicating their applicability in practice.

We also derive insights into the HSCP under different settings. In particular, we show how one can use the proposed models to obtain staffing and capacity allocation decisions under different cost structures that the decision-maker can easily change in the models. Our results also show how different methods of modeling uncertainty in the HSCP lead to different staffing and capacity allocation decisions with varying performance. These results address the primary objective of this paper. That is, to investigate the value of SP and DRO approaches for this specific HSCP problem. Our analyses and results could help practitioners decide which approach to implement in practice to make staffing and capacity planning decisions based on their preferences and practical settings. More broadly, our results motivate the need for considering different approaches when modeling uncertainty and draw attention to the need to model the distributional ambiguity of uncertain problem data in strategic real-world stochastic optimization problems such as the HSCP problem. Finally, we remark that our DRO models for the EA decision-maker have the advantage of being directly implementable, i.e., they can be implemented and solved using standard optimization software. In contrast, the DRO models for the FA decision-maker require specialized algorithms.

We suggest the following areas for future research. First, the proposed approaches can be extended to incorporate other multi-modal sources of uncertainty (e.g., travel time). Second, our models can be considered as the first step toward building data-driven and robust home service planning including designing routing plans for providers and assigning appointment times for customers, among other important considerations. Finally, it will be interesting and relevant to practice to incorporate other aspects, such as the possibility of hiring part-time providers and the unexpected availability of the hired full-time providers on the day of service.

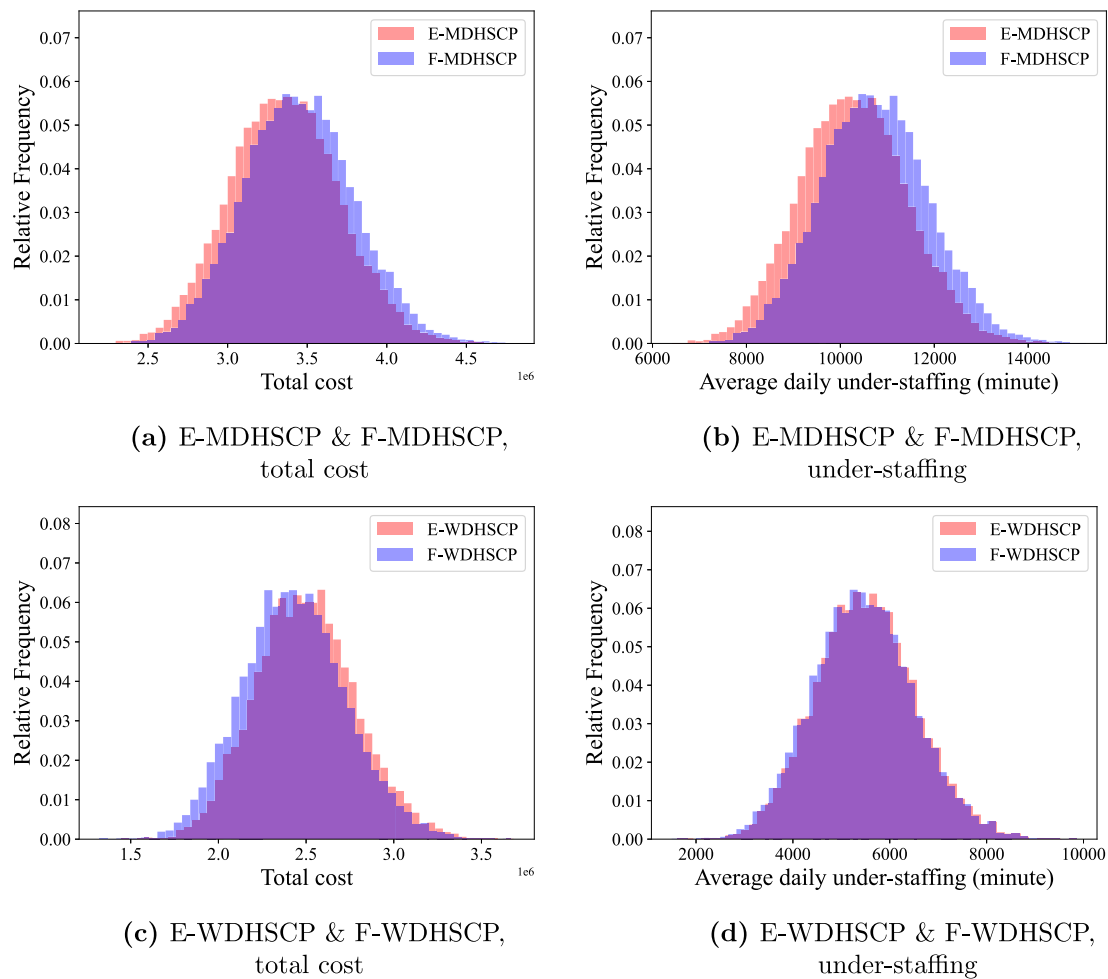


Fig. 12. Out-of-sample of EA models and FA models under Set 2, $\Delta = 0$.

CRedit authorship contribution statement

Ridong Wang: Conceptualization, Formal analysis, Methodology, Computational investigation, Computational insights, Writing – original draft, Writing – review & editing. **Karmel S. Shehadeh:** Conceptualization, Formal analysis, Methodology, Computational insights, Writing – original draft, Writing – review & editing. **Xiaolei Xie:** Conceptualization, Formal analysis, Methodology, Computational insights, Writing – review & editing. **Lefei Li:** Conceptualization, Formal analysis, Methodology, Computational insights, Writing – review & editing.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cor.2023.106348>.

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